US 20250287395A1

(19) United States

(12) Patent Application Publication

Xiong et al.

(10) Pub. No.: US 2025/0287395 A1

(43) Pub. Date: Sep. 11, 2025

(54) MODE 1 RESOURCE ALLOCATION OF POSITIONING REFERENCE SIGNALS

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **Gang Xiong**, Beaverton, OR (US); **Debdeep Chatterjee**, San Jose, CA (US); **Kilian Peter Anton Roth**, München (DE); **Toufiqul Islam**, Santa Clara, CA (US); **Jihyun Lee**, Santa Clara, CA (US)

(21) Appl. No.: **18/858,547**

(22) PCT Filed: **Jul. 6, 2023**

(86) PCT No.: **PCT/US2023/027023**

§ 371 (c)(1), (2) Date: **Oct. 21, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/391,661, filed on Jul. 22, 2022, provisional application No. 63/501,495, filed on May 11, 2023.

Publication Classification

(51) **Int. Cl.**

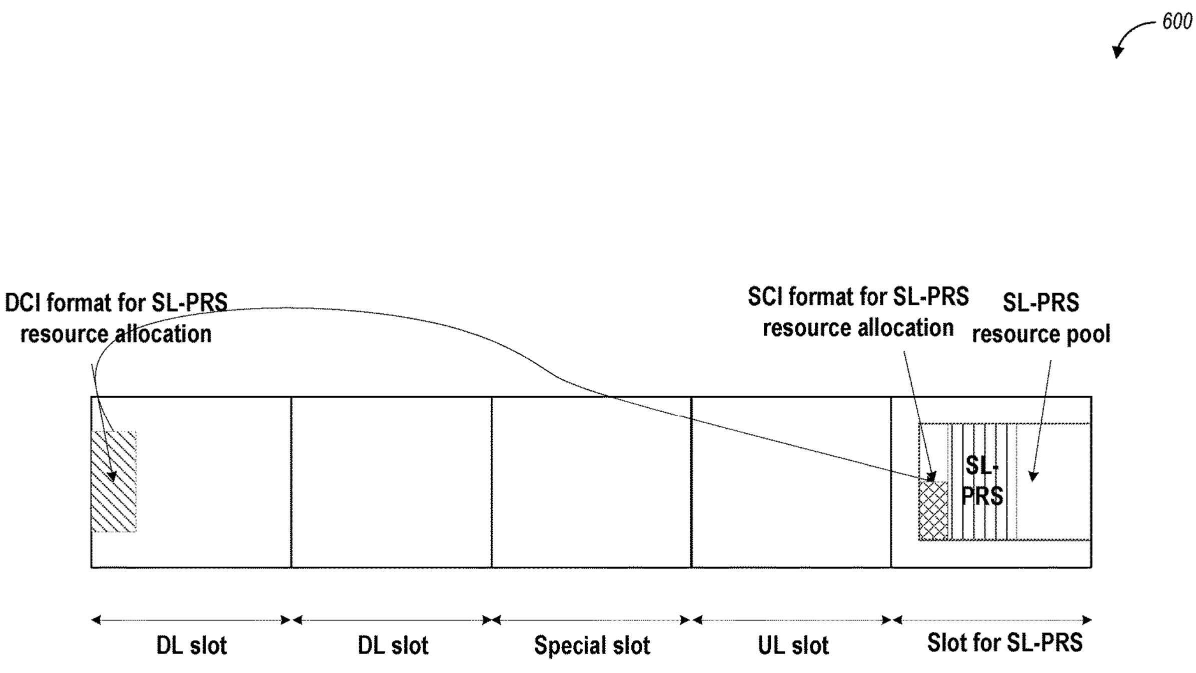
H04W 72/232	(2023.01)
H04L 5/00	(2006.01)
H04W 64/00	(2009.01)
H04W 72/25	(2023.01)

(52) **U.S. Cl.**

CPC ***H04W 72/232*** (2023.01); ***H04L 5/0053*** (2013.01); ***H04L 5/0094*** (2013.01); ***H04W 64/00*** (2013.01); ***H04W 72/25*** (2023.01)

(57) **ABSTRACT**

A computer-readable storage medium stores instructions to configure a base station for mode 1 resource allocation for sidelink (SL) positioning in a 5G NR network, and to cause the UE to perform operations including decoding RRC signaling, LTE positioning protocol (EPP) signaling received from a base station for an in-coverage scenario, and/or sidelink positioning protocol (SEPP) signaling from a second UE. The EPP, RRC, or SEPP signaling respectively includes configuration information configuring a resource pool associated with positioning reference signal communications. The DCI is decoded from the base station using a PDCCH. The DCI indicates a resource pool index and/or the presence of a sidelink positioning reference signal (SL PRS) in the indicated resource pool. An SL PRS is encoded for transmission to the second UE using the indicated resource pool.



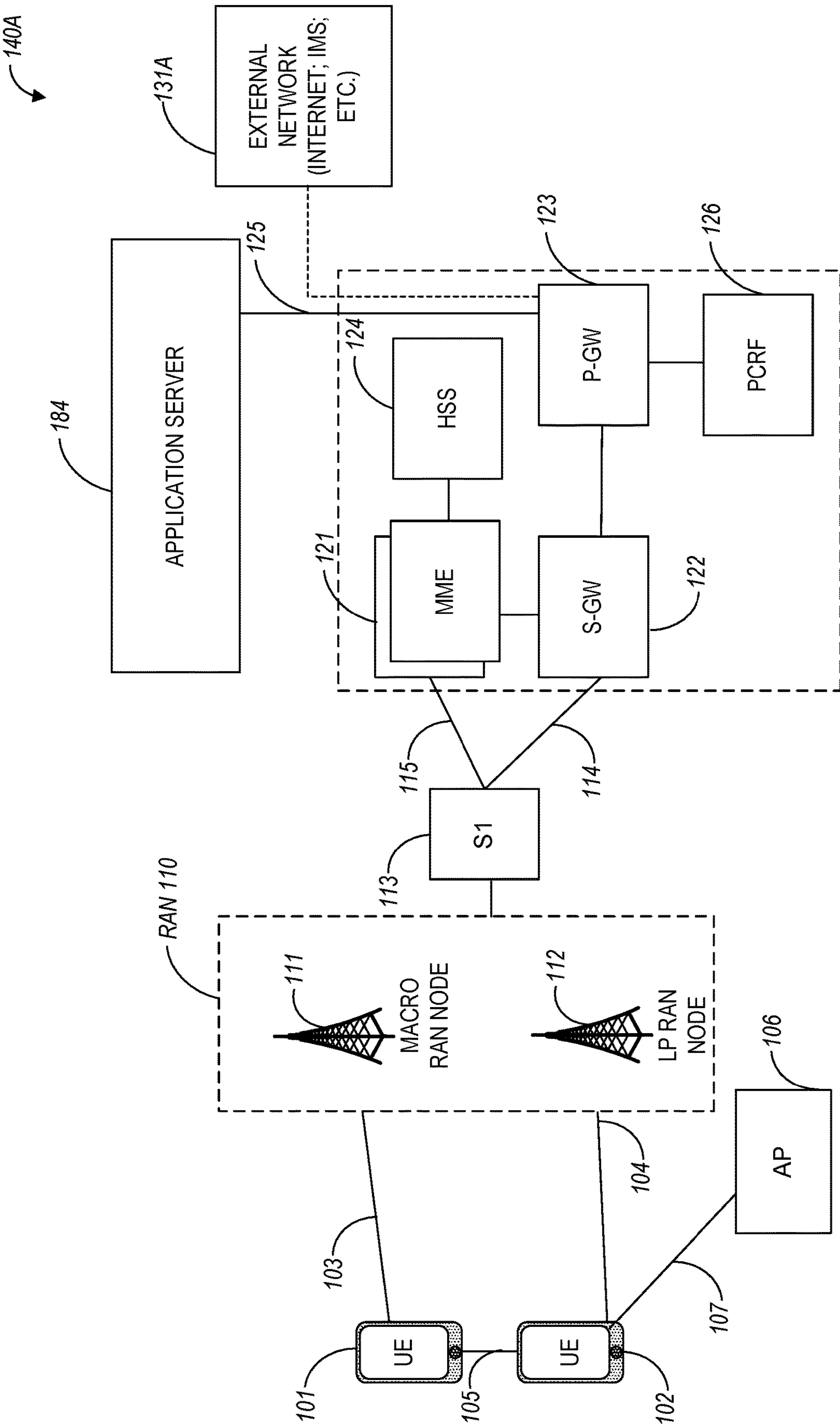

140A
EXTERNAL NETWORK (INTERNET; IMS; ETC.)
131A
APPLICATION SERVER
184
125
123
126
124
HSS
P-GW
PCRF
MME
S-GW
121
122
CN (EPC) 120
115
114
S1
113
RAN 110
111
MACRO RAN NODE
112
LP RAN NODE
106
AP
104
103
107
UE
UE
101
105
102

FIG. 1A

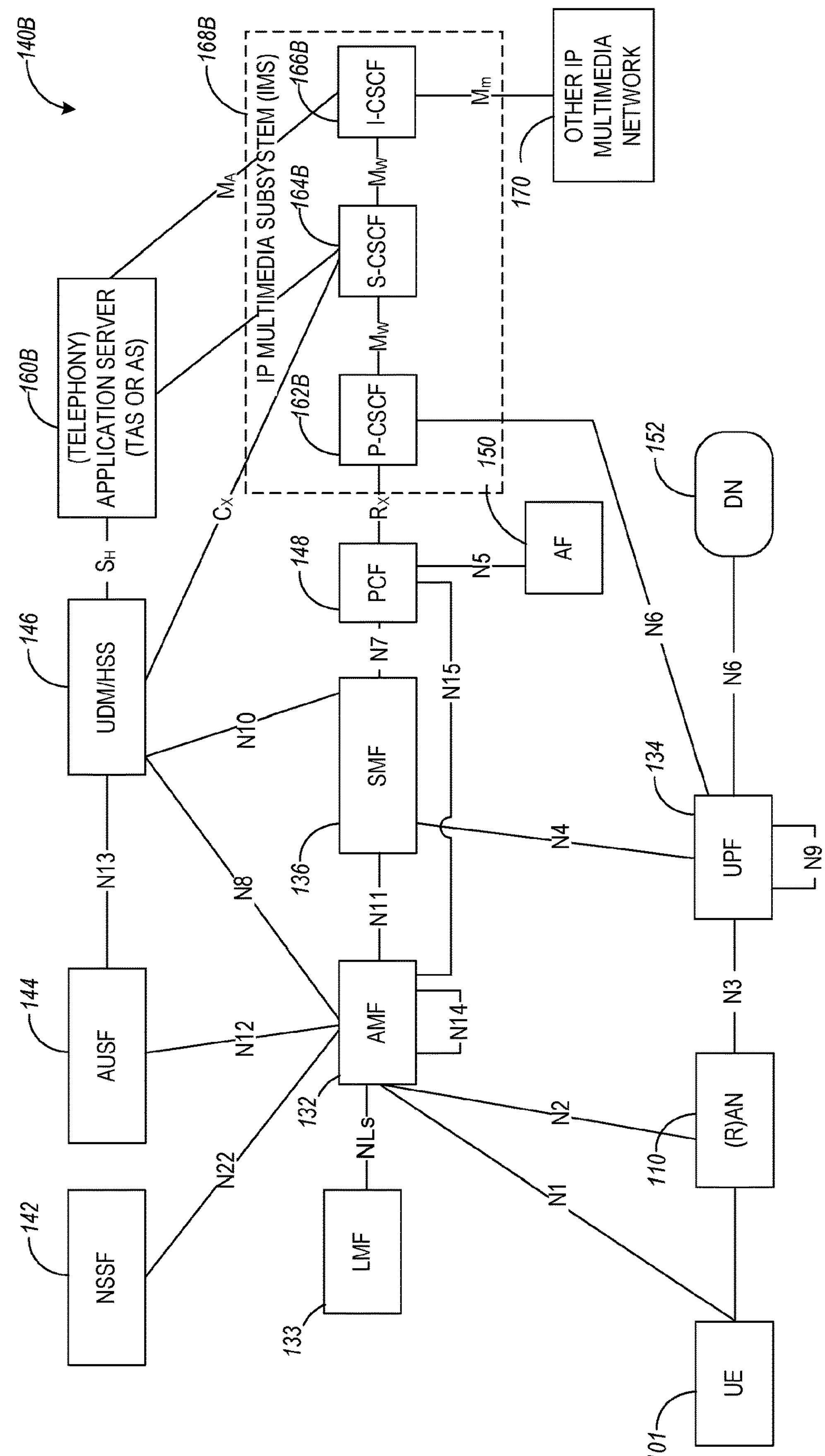

140B
168B
IP MULTIMEDIA SUBSYSTEM (IMS)
166B
I-CSCF
164B
S-CSCF
162B
P-CSCF
M_W
M_W
M_m
M_A
OTHER IP MULTIMEDIA NETWORK
170
160B
(TELEPHONY) APPLICATION SERVER (TAS OR AS)
S_H
C_X
R_X
146
UDM/HSS
148
PCF
150
AF
N5
152
DN
N6
N6
N7
N15
N10
SMF
136
N4
134
UPF
N9
N13
N8
N11
144
AUSF
N12
AMF
N14
N3
132
NLs
N2
110
(R)AN
142
NSSF
N22
LMF
133
N1
UE
101

FIG. 1B

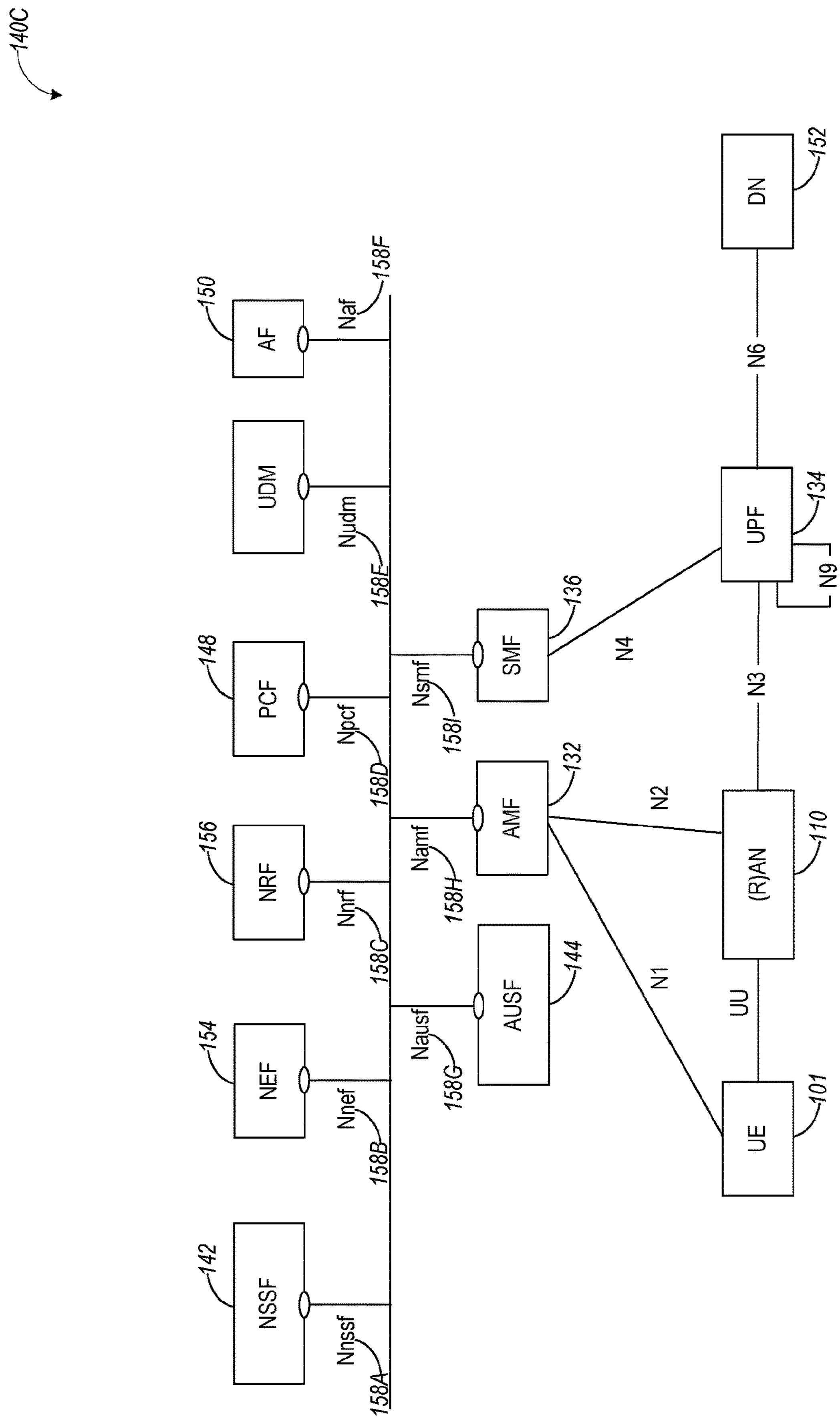

140C
NSSF
142
Nnssf
158A
NEF
154
Nnef
158B
NRF
156
Nnrf
158C
PCF
148
Npcf
158D
UDM
Nudm
158E
AF
150
Naf
158F
AUSF
144
Nausf
158G
AMF
132
Namf
158H
SMF
136
Nsmf
158I
UE
101
UU
N1
(R)AN
110
N2
N3
N4
UPF
134
N9
N6
DN
152

FIG. 1C

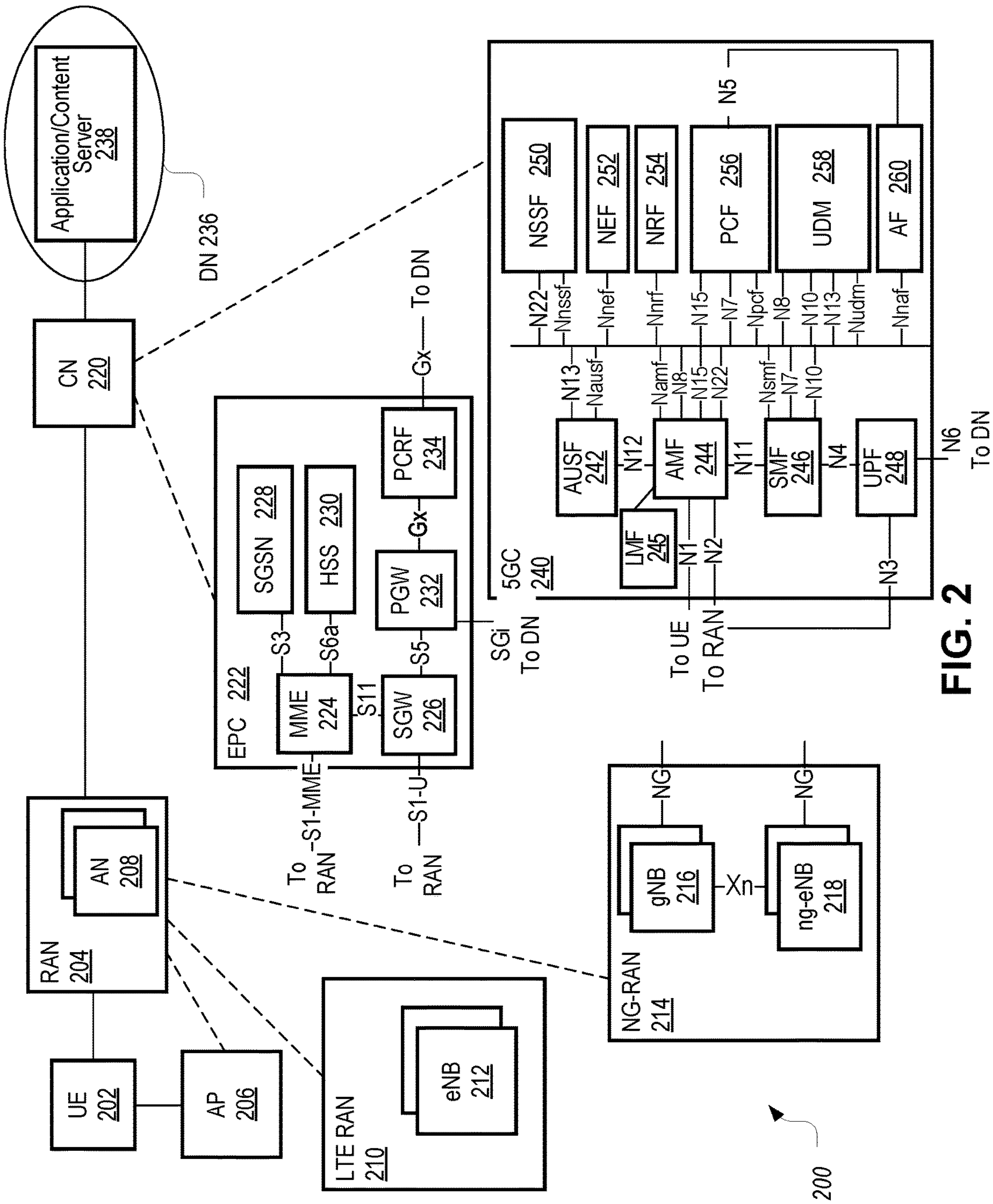

UE 202
AP 206
RAN 204
AN 208
LTE RAN 210
eNB 212
NG-RAN 214
gNB 216
ng-eNB 218
Xn
NG
CN 220
DN 236
Application/Content Server 238
EPC 222
MME 224
SGW 226
SGSN 228
HSS 230
PGW 232
PCRF 234
S3
S6a
S11
S5
Gx
To DN
SGi To DN
To RAN
S1-MME
S1-U
5GC 240
AUSF 242
AMF 244
LMF 245
SMF 246
UPF 248
NSSF 250
NEF 252
NRF 254
PCF 256
UDM 258
AF 260
N22
Nnssf
Nnef
Nnrf
N15
N7
Npcf
N8
N10
N13
Nudm
Nnaf
Nausf
Namf
Nsmf
N12
N11
N4
N6 To DN
N5
N1
N2
N3
To UE
To RAN
200

FIG. 2

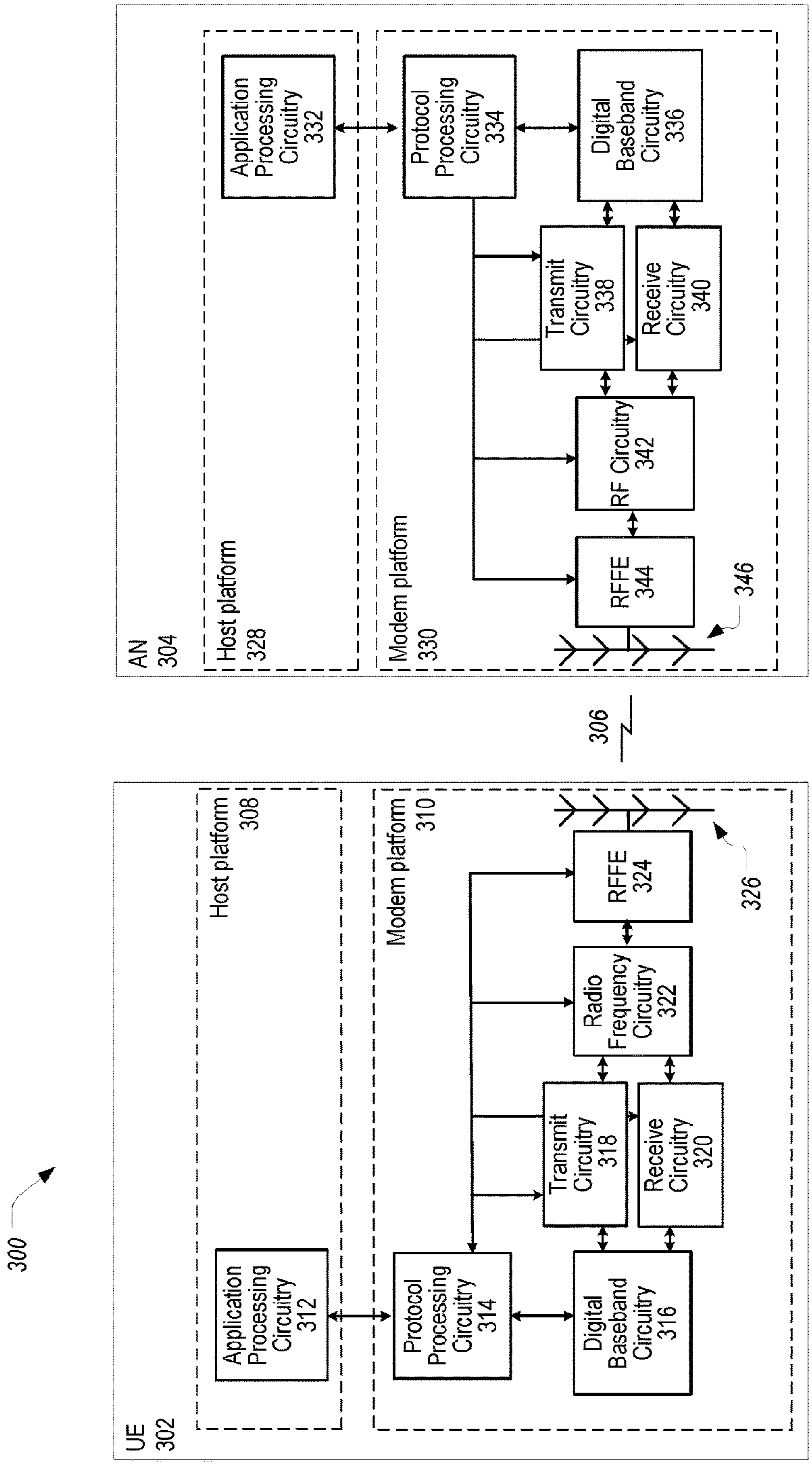

300
UE
302
Host platform
308
Application Processing Circuitry
312
Modem platform
310
Protocol Processing Circuitry
314
Digital Baseband Circuitry
316
Transmit Circuitry
318
Receive Circuitry
320
Radio Frequency Circuitry
322
RFFE
324
326
306
AN
304
Host platform
328
Application Processing Circuitry
332
Modem platform
330
Protocol Processing Circuitry
334
Digital Baseband Circuitry
336
Transmit Circuitry
338
Receive Circuitry
340
RF Circuitry
342
RFFE
344
346

FIG. 3

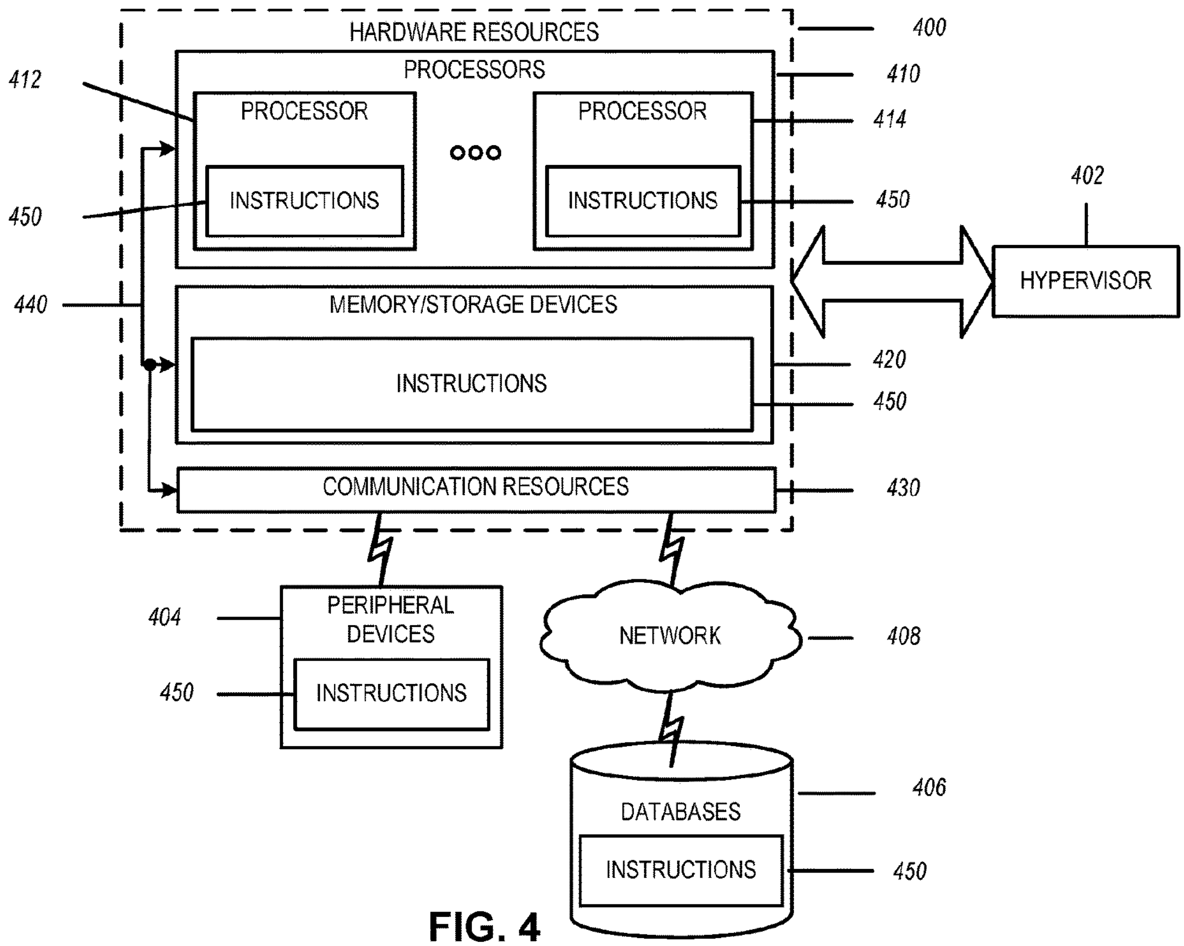

HARDWARE RESOURCES
400
PROCESSORS
410
412
PROCESSOR
PROCESSOR
414
INSTRUCTIONS
INSTRUCTIONS
450
450
402
HYPERVISOR
440
MEMORY/STORAGE DEVICES
INSTRUCTIONS
420
450
COMMUNICATION RESOURCES
430
404
PERIPHERAL DEVICES
450
INSTRUCTIONS
NETWORK
408
406
DATABASES
INSTRUCTIONS
450

FIG. 4

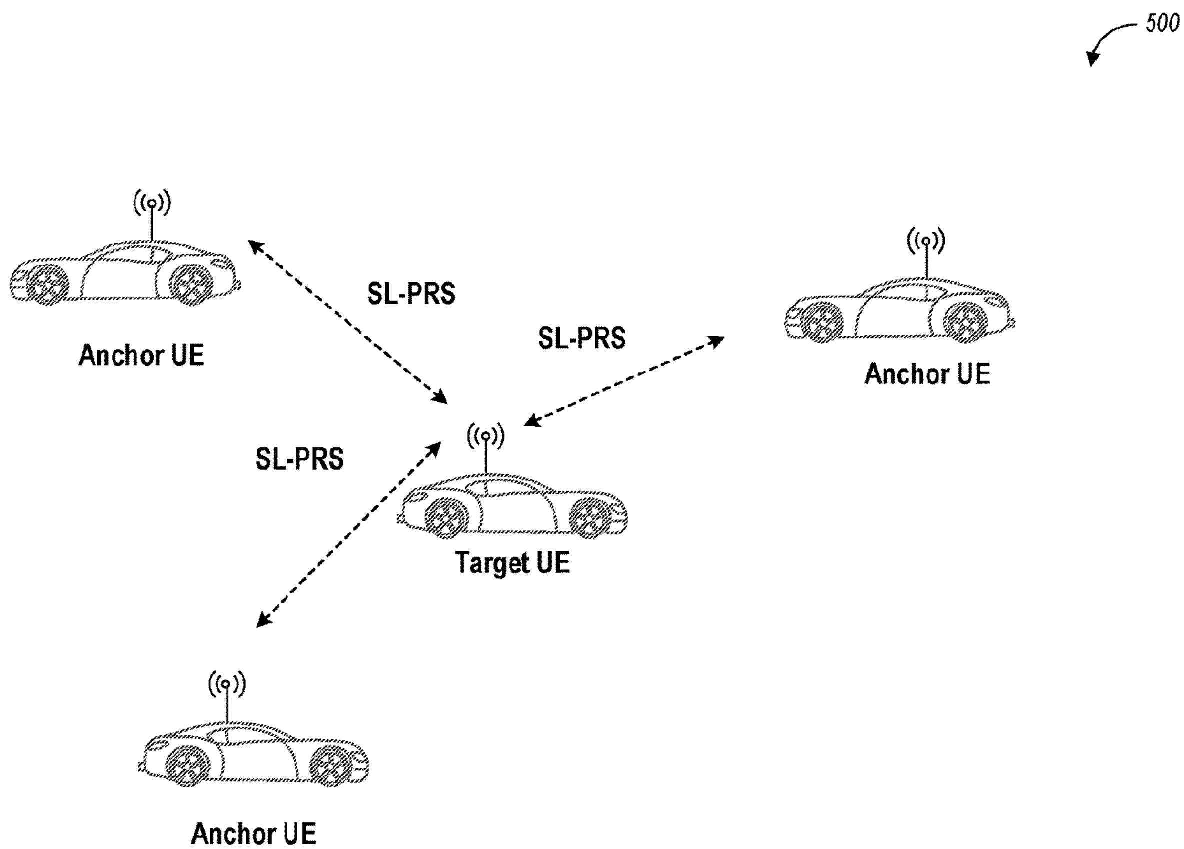

500
SL-PRS
SL-PRS
SL-PRS
Anchor UE
Anchor UE
Target UE
Anchor UE

FIG. 5

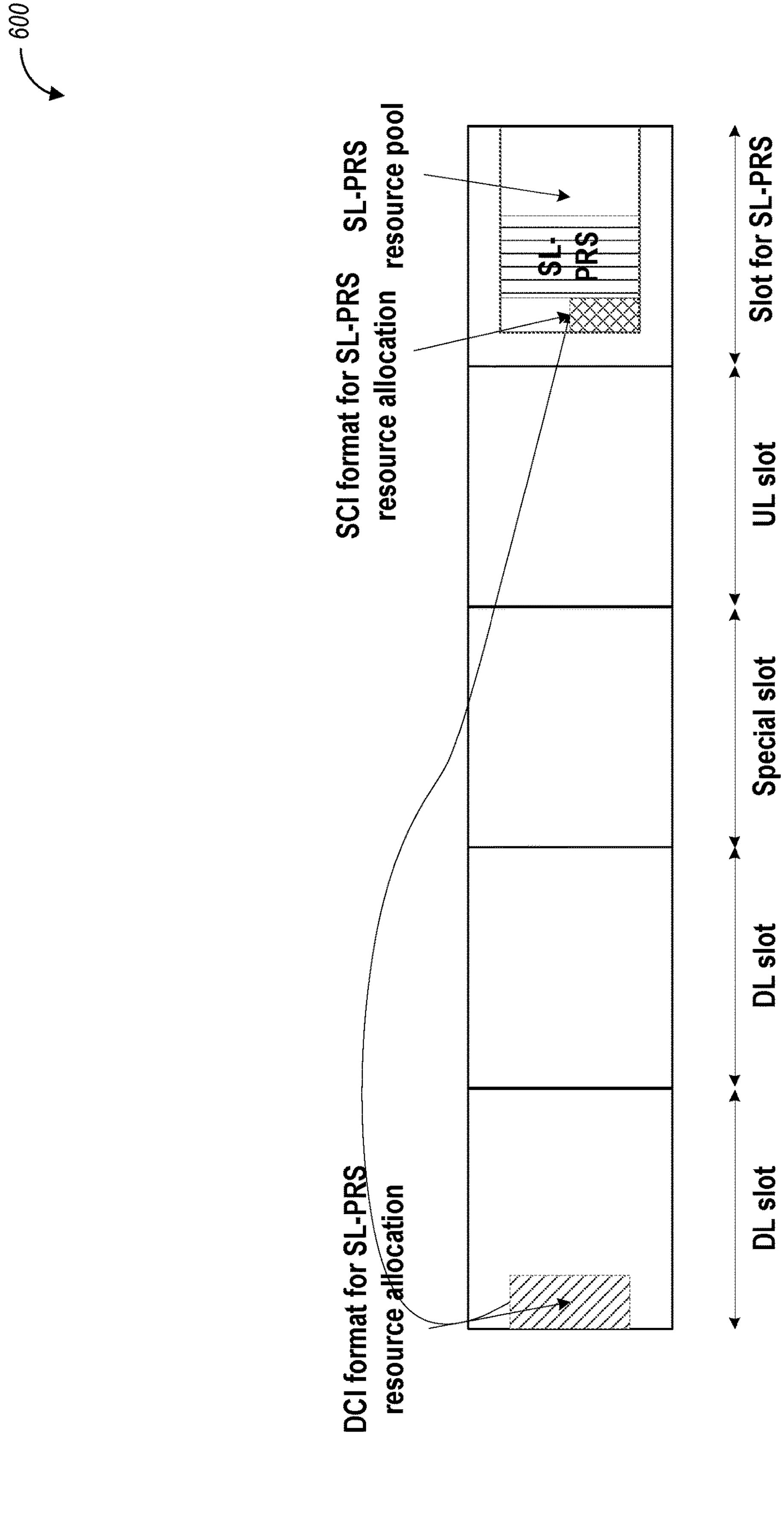

600
SL-PRS resource pool
SCI format for SL-PRS resource allocation
SL-PRS
DCI format for SL-PRS resource allocation
Slot for SL-PRS
UL slot
Special slot
DL slot
DL slot

FIG. 6

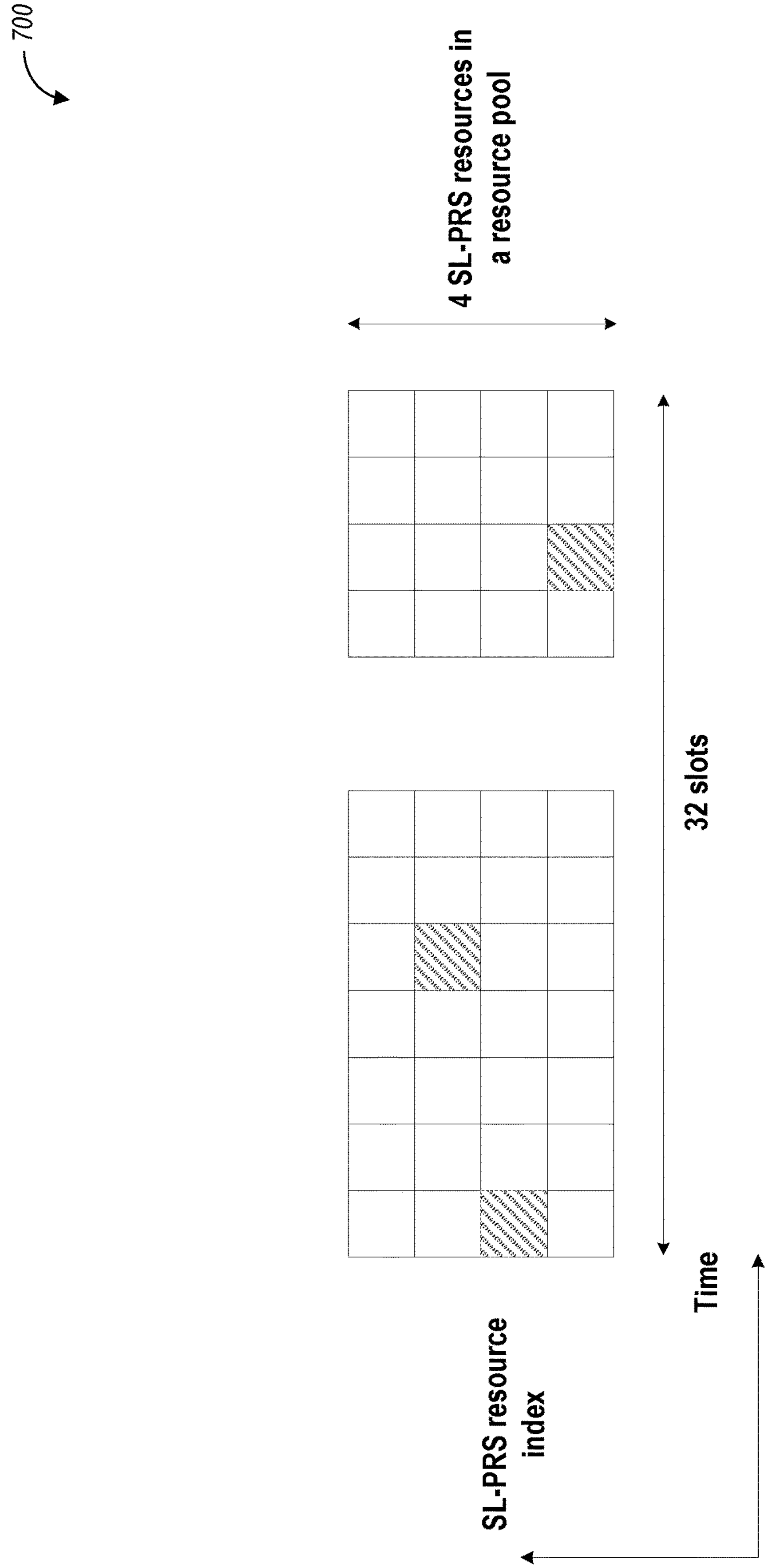

700
4 SL-PRS resources in a resource pool
32 slots
Time
SL-PRS resource index

FIG. 7

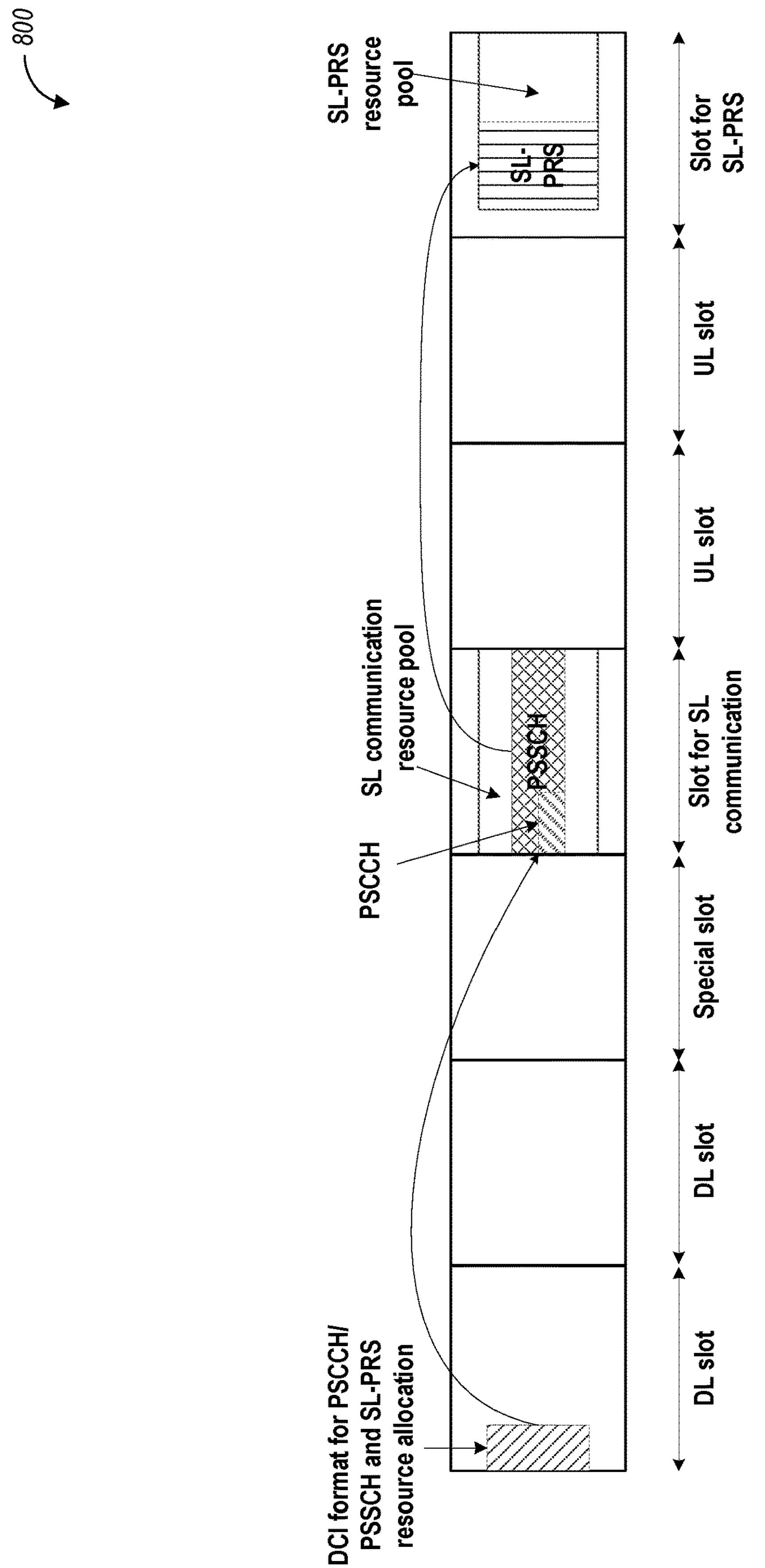
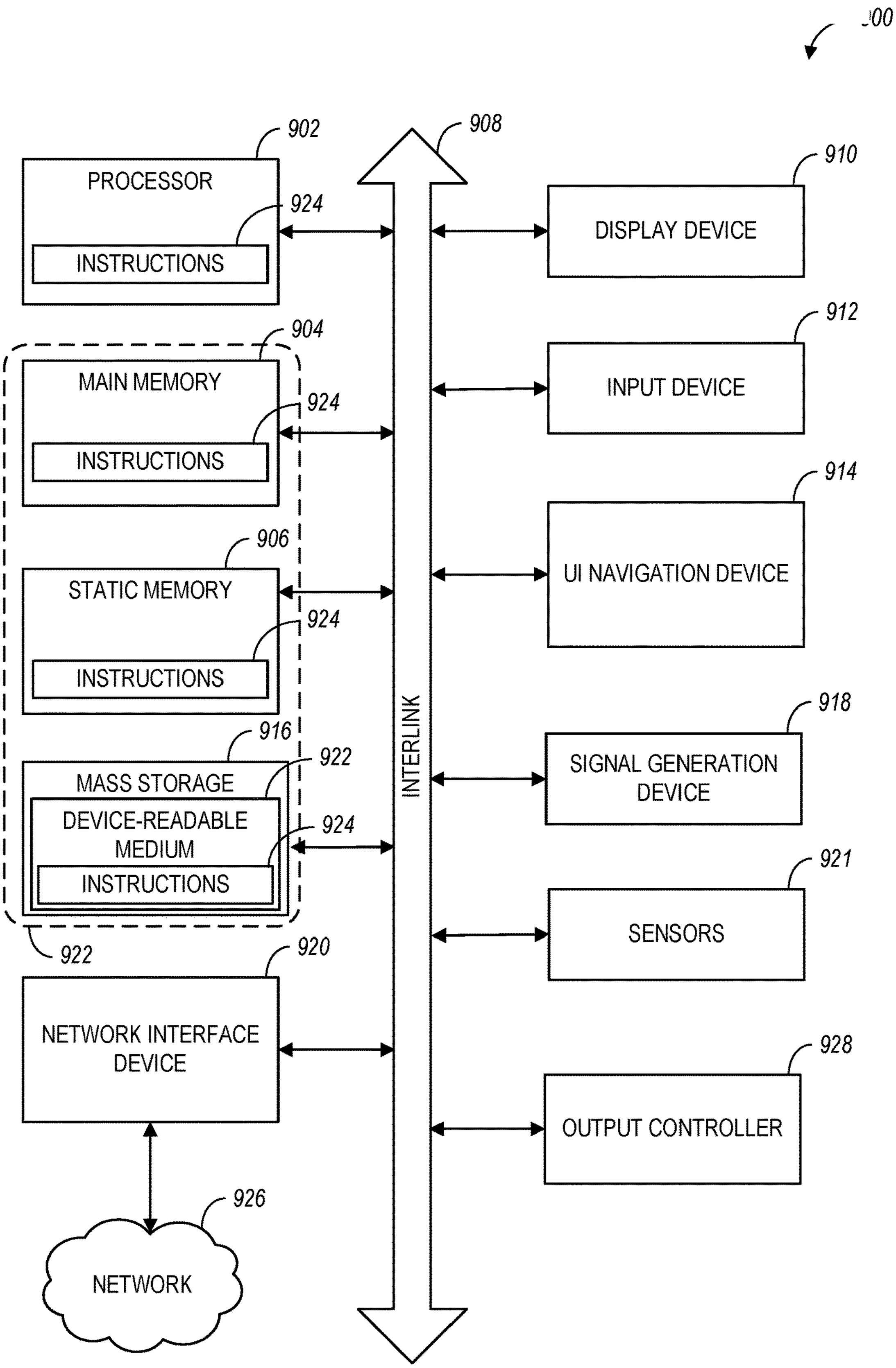

800
SL-PRS resource pool
SL-PRS
Slot for SL-PRS
UL slot
UL slot
SL communication resource pool
PSSCH
PSCCH
Slot for SL communication
Special slot
DL slot
DL slot
DCI format for PSCCH/ PSSCH and SL-PRS resource allocation

FIG. 8



900
902
PROCESSOR
924
INSTRUCTIONS
904
MAIN MEMORY
924
INSTRUCTIONS
906
STATIC MEMORY
924
INSTRUCTIONS
916
MASS STORAGE
922
DEVICE-READABLE MEDIUM
924
INSTRUCTIONS
922
920
NETWORK INTERFACE DEVICE
926
NETWORK
908
INTERLINK
910
DISPLAY DEVICE
912
INPUT DEVICE
914
UI NAVIGATION DEVICE
918
SIGNAL GENERATION DEVICE
921
SENSORS
928
OUTPUT CONTROLLER

FIG. 9

MODE 1 RESOURCE ALLOCATION OF POSITIONING REFERENCE SIGNALS

PRIORITY CLAIM

[0001] This application claims the benefit of priority to the following patent applications:

[0002] U.S. Provisional Patent Application No. 63/391,661, filed Jul. 22, 2022, and entitled "SYSTEMS AND METHODS ON MODE 1 RESOURCE ALLOCATION OF POSITIONING REFERENCE SIGNAL FOR NR SIDELINK;" and

[0003] U.S. Provisional Patent Application No. 63/501,495, filed May 11, 2023, and entitled "SYSTEMS AND METHODS ON MODE 1 RESOURCE ALLOCATION OF POSITIONING REFERENCE SIGNAL FOR NR SIDELINK."

[0004] Each of the above-listed applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0005] Aspects pertain to wireless communications. Some aspects relate to wireless networks including 3GPP (Third Generation Partnership Project) networks, 3GPP LTE (Long Term Evolution) networks, 3GPP LTE-A (LTE Advanced) networks, (MulteFire, LTE-U), and fifth-generation (5G) networks including 5G new radio (NR) (or 5G-NR) networks, 5G-LTE networks such as 5G NR unlicensed spectrum (NR-U) networks and other unlicensed networks including Wi-Fi, CBRS (OnGo), etc. Other aspects are directed to techniques for configuring mode 1 resource allocation of positioning reference signals for sidelink (SL) communications.

BACKGROUND

[0006] Mobile communications have evolved significantly from early voice systems to today's highly sophisticated integrated communication platform. With the increase in different types of devices communicating with various network devices, the usage of 3GPP LTE systems has increased. The penetration of mobile devices (user equipment or UEs) in modern society has continued to drive demand for a wide variety of networked devices in many disparate environments. Fifth-generation (5G) wireless systems are forthcoming and are expected to enable even greater speed, connectivity, and usability. Next-generation 5G networks (or NR networks) are expected to increase throughput, coverage, and robustness and reduce latency and operational and capital expenditures. 5G-NR networks will continue to evolve based on 3GPP LTE-Advanced with additional potential new radio access technologies (RATs) to enrich people's lives with seamless wireless connectivity solutions delivering fast, rich content and services. As the current cellular network frequency is saturated, higher frequencies, such as millimeter wave (mmWave) frequency, can be beneficial due to their high bandwidth.

[0007] Potential LTE operation in the unlicensed spectrum includes (and is not limited to) the LTE operation in the unlicensed spectrum via dual connectivity (DC), or DC-based LAA, and the standalone LTE system in the unlicensed spectrum, according to which LTE-based technology solely operates in the unlicensed spectrum without requiring an "anchor" in the licensed spectrum, called MulteFire. Further enhanced operation of LTE and NR systems in the licensed, as well as unlicensed spectrum, is expected in future releases and 5G systems. Such enhanced operations can include techniques for configuring mode 1 resource allocation of positioning reference signals for sidelink (SL) communications.

BRIEF DESCRIPTION OF THE FIGURES

[0008] In the figures, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The figures illustrate generally, by way of example, but not by way of limitation, various aspects discussed in the present document.

[0009] FIG. **1**A illustrates an architecture of a network, in accordance with some aspects.

[0010] FIG. **1**B and FIG. **1**C illustrate a non-roaming 5G system architecture in accordance with some aspects.

[0011] FIG. **2**, FIG. **3**, and FIG. **4** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

[0012] FIG. **5** illustrates an example of sidelink positioning with anchor UEs and a target UE, in accordance with some aspects.

[0013] FIG. **6** illustrates a diagram of a time gap between a downlink control information (DCI) format, a sidelink control information (SCI) format, and a sidelink positioning reference signal (SL PRS) transmission, in accordance with some aspects.

[0014] FIG. **7** illustrates a diagram of resource allocation and reservation for a SL PRS transmission, in accordance with some aspects.

[0015] FIG. **8** illustrates a diagram of a time gap between a DCI format, a PSCCH/PSSCH transmission, and an SL PRS transmission, in accordance with some aspects.

[0016] FIG. **9** illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node), an NCR, an access point (AP), a wireless station (STA), a mobile station (MS), or user equipment (UE), in accordance with some aspects.

DETAILED DESCRIPTION

[0017] The following description and the drawings sufficiently illustrate aspects to enable those skilled in the art to practice them. Other aspects may incorporate structural, logical, electrical, process, and other changes. Portions and features of some aspects may be included in or substituted for, those of other aspects. Aspects outlined in the claims encompass all available equivalents of those claims.

[0018] FIG. **1**A illustrates an architecture of a network in accordance with some aspects. The communication network **140**A is shown to include user equipment (UE) **101** and UE **102**. The UE **101** and UE **102** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks) but may also include any mobile or non-mobile computing device, such as Personal Data Assistants (PDAs), pagers, laptop computers, desktop computers, wireless handsets, drones, or any other computing device including a wired and/or wireless communications interface. UE **101** and UE **102** can be collectively referred to herein as UE **101**, and UE **101** can be used to perform one or more of the techniques disclosed herein.

[0019] Any of the radio links described herein (e.g., as used in the network **140**A or any other illustrated network) may operate according to any exemplary radio communication technology and/or standard.

[0020] LTE and LTE-Advanced are standards for wireless communications of high-speed data for UE such as mobile telephones. In LTE-Advanced and various wireless systems, carrier aggregation is a technology according to which multiple carrier signals operating on different frequencies may be used to carry communications for a single UE, thus increasing the bandwidth available to a single device. In some aspects, carrier aggregation may be used where one or more component carriers operate on unlicensed frequencies.

[0021] Aspects described herein can be used in the context of any spectrum management scheme including, for example, dedicated licensed spectrum, unlicensed spectrum, (licensed) shared spectrum (such as Licensed Shared Access (LSA) in 2.3-2.4 GHz, 3.4-3.6 GHz, 3.6-3.8 GHz, and further frequencies and Spectrum Access System (SAS) in 3.55-3.7 GHZ and further frequencies).

[0022] Aspects described herein can also be applied to different Single Carrier or OFDM flavors (CP-OFDM, SC-FDMA, SC-OFDM, filter bank-based multicarrier (FBMC), OFDMA, etc.) and in particular 3GPP NR (New Radio) by allocating the OFDM carrier data bit vectors to the corresponding symbol resources.

[0023] In some aspects, any of the UE **101** and UE **102** can comprise an Internet-of-Things (IoT) UE or a Cellular IoT (CIoT) UE, which can comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. In some aspects, any of the UE **101** and UE **102** can include a narrowband (NB) IoT UE (e.g., such as an enhanced NB-IoT (eNB-IoT) UE and Further Enhanced (FeNB-IoT) UE). An IoT UE can utilize technologies such as machine-to-machine (M2M) or machine-type communications (MTC) for exchanging data with an MTC server or device via a public land mobile network (PLMN), Proximity-Based Service (ProSe), or device-to-device (D2D) communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network includes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

[0024] In some aspects, any of the UE **101** and UE **102** can include enhanced MTC (eMTC) UEs or further enhanced MTC (FeMTC) UEs.

[0025] The UE **101** and UE **102** may be configured to connect, e.g., communicatively coupled, with a radio access network (RAN) **110**. The RAN **110** may be, for example, a Universal Mobile Telecommunications System (UMTS), an Evolved Universal Terrestrial Radio Access Network (E-UTRAN), a NextGen RAN (NG RAN), or some other type of RAN. The UE **101** and UE **102** utilize connections **103** and **104**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below); in this example, the connections **103** and **104** are illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols, such as a Global System for Mobile Communications (GSM) protocol, a code-division multiple access (CDMA) network protocol, a Push-to-Talk (PTT) protocol, a PTT over Cellular (POC) protocol, a Universal Mobile Telecommunications System (UMTS) protocol, a 3GPP Long Term Evolution (LTE) protocol, a fifth-generation (5G) protocol, a New Radio (NR) protocol, and the like.

[0026] In an aspect, the UE **101** and UE **102** may further directly exchange communication data via a ProSe interface **105**. The ProSe interface **105** may alternatively be referred to as a sidelink interface comprising one or more logical channels, including but not limited to a Physical Sidelink Control Channel (PSCCH), a Physical Sidelink Shared Channel (PSSCH), a Physical Sidelink Discovery Channel (PSDCH), and a Physical Sidelink Broadcast Channel (PSBCH).

[0027] The UE **102** is shown to be configured to access an access point (AP) **106** via connection **107**. Connection **107** can comprise a local wireless connection, such as, for example, a connection consistent with any IEEE 802.11 protocol, according to which the AP **106** can comprise a wireless fidelity (WiFi®) router. In this example, the AP **106** is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below).

[0028] The RAN **110** can include one or more access nodes that enable connections **103** and **104**. These access nodes (ANs) can be referred to as base stations (BSs), NodeBs, evolved NodeBs (eNBs), Next Generation NodeBs (gNBs), RAN network nodes, and the like, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). In some aspects, communication nodes **111** and **112** can be transmission/reception points (TRPs). In instances when the communication nodes **111** and **112** are NodeBs (e.g., eNBs or gNBs), one or more TRPs can function within the communication cell of the NodeBs. The RAN **110** may include one or more RAN nodes for providing macrocells, e.g., macro RAN node **111**, and one or more RAN nodes for providing femtocells or picocells (e.g., cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells), e.g., low power (LP) RAN node **112** or an unlicensed spectrum based secondary RAN node **112**.

[0029] Any of the RAN nodes **111** and **112** can terminate the air interface protocol and can be the first point of contact for UE **101** and UE **102**. In some aspects, any of the RAN nodes **111** and **112** can fulfill various logical functions for the RAN **110** including, but not limited to, the radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling, and mobility management. In an example, any of the nodes **111** and/or **112** can be a new generation Node-B (gNB), an evolved node-B (eNB), or another type of RAN node.

[0030] The RAN **110** is shown to be communicatively coupled to a core network (CN) **120** via an S1 interface **113**. In aspects, the CN **120** may be an evolved packet core (EPC) network, a NextGen Packet Core (NPC) network, or some other type of CN (e.g., as illustrated in FIGS. **1B**-**1C**). In this aspect, the S1 interface **113** is split into two parts: the S1-U interface **114**, which carries user traffic data between the RAN nodes **111** and **112** and the serving gateway (S-GW)

122, and the S1-mobility management entity (MME) interface **115**, which is a signaling interface between the RAN nodes **111** and **112** and MMEs **121**.

[0031] In this aspect, the CN **120** comprises the MMEs **121**, the S-GW **122**, the Packet Data Network (PDN) Gateway (P-GW) **123**, and a home subscriber server (HSS) **124**. The MMEs **121** may be similar in function to the control plane of legacy Serving General Packet Radio Service (GPRS) Support Nodes (SGSN). The MMEs **121** may manage mobility aspects in access such as gateway selection and tracking area list management. The HSS **124** may comprise a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The CN **120** may comprise one or several HSSs **124**, depending on the number of mobile subscribers, the capacity of the equipment, the organization of the network, etc. For example, the HSS **124** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc.

[0032] The S-GW **122** may terminate the S1 interface **113** towards the RAN **110**, and route data packets between the RAN **110** and the CN **120**. In addition, the S-GW **122** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities of the S-GW **122** may include lawful intercept, charging, and some policy enforcement.

[0033] The P-GW **123** may terminate an SGi interface toward a PDN. The P-GW **123** may route data packets between the EPC network (e.g., CN **120**) and external networks such as a network including the application server **184** (alternatively referred to as application function (AF)) via an Internet Protocol (IP) interface **125**. The P-GW **123** can also communicate data to other external networks **131**A, which can include the Internet, IP multimedia subsystem (IPS) network, and other networks. Generally, the application server **184** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS Packet Services (PS) domain, LTE PS data services, etc.). In this aspect, the P-GW **123** is shown to be communicatively coupled to an application server **184** via an IP interface **125**. The application server **184** can also be configured to support one or more communication services (e.g., Voice-over-Internet Protocol (VOIP) sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UE **101** and UE **102** via the CN **120**. The P-GW **123** may further be a node for policy enforcement and

[0034] charging data collection. Policy and Charging Rules Function (PCRF) **126** is the policy and charging control element of the CN **120**. In a non-roaming scenario, in some aspects, there may be a single PCRF in the Home Public Land Mobile Network (HPLMN) associated with a UE's Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with a local breakout of traffic, there may be two PCRFs associated with a UE's IP-CAN session: a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **126** may be communicatively coupled to the application server **184** via the P-GW **123**.

[0035] In some aspects, the communication network **140**A can be an IoT network or a 5G network, including a 5G new radio network using communications in the licensed (5G NR) and the unlicensed (5G NR-U) spectrum. One of the current enablers of IoT is the narrowband IoT (NB-IoT).

[0036] An NG system architecture can include the RAN **110** and a 5G core network (e.g., CN **120**). RAN **110** in an NG system can be referred to as NG-RAN. The NG-RAN can include a plurality of nodes, such as gNBs and NG-eNBs. The CN **120** (also referred to as a 5G core network or 5GC) can include an access and mobility function (AMF) and/or a user plane function (UPF). The AMF and the UPF can be communicatively coupled to the gNBs and the NG-eNBs via NG interfaces. More specifically, in some aspects, the gNBs and the NG-eNBs can be connected to the AMF by NG-C interfaces, and the UPF by NG-U interfaces. The gNBs and the NG-eNBs can be coupled to each other via Xn interfaces.

[0037] In some aspects, the NG system architecture can use reference points between various nodes as provided by 3GPP Technical Specification (TS) 23.501 (e.g., V 15.4.0, 2018 December). In some aspects, each of the gNBs and the NG-eNBs can be implemented as a base station, a mobile edge server, a small cell, a home eNB, a RAN network node, and so forth. In some aspects, a gNB can be a master node (MN) and NG-eNB can be a secondary node (SN) in a 5G architecture. In some aspects, the master/primary node may operate in a licensed band and the secondary node may operate in an unlicensed band.

[0038] FIG. **1**B illustrates a non-roaming 5G system architecture in accordance with some aspects. Referring to FIG. **1**B, there is illustrated a 5G system architecture **140**B in a reference point representation. More specifically, UE **102** can be in communication with RAN **110** as well as one or more other 5G core (5GC) network entities. The 5G system architecture **140**B includes a plurality of network functions (NFs), such as access and mobility management function (AMF) **132**, location management function (LMF) **133**, session management function (SMF) **136**, policy control function (PCF) **148**, application function (AF) **150**, user plane function (UPF) **134**, network slice selection function (NSSF) **142**, authentication server function (AUSF) **144**, and unified data management (UDM)/home subscriber server (HSS) **146**. The UPF **134** can provide a connection to a data network (DN) **152**, which can include, for example, operator services, Internet access, or third-party services. The AMF **132** can be used to manage access control and mobility and can also include network slice selection functionality. The SMF **136** can be configured to set up and manage various sessions according to network policy. The UPF **134** can be deployed in one or more configurations according to the desired service type. The PCF **148** can be configured to provide a policy framework using network slicing, mobility management, and roaming (similar to PCRF in a 4G communication system). The UDM can be configured to store subscriber profiles and data (similar to an HSS in a 4G communication system).

[0039] The LMF **133** may be used in connection with 5G positioning functionalities. In some aspects, LMF **133** receives measurements and assistance information from RAN **110** (which can be NG-RAN) and the mobile device (e.g., UE **101**) via the AMF **132** over the NLs interface to compute the position of the UE **101**. In some aspects, NR positioning protocol A (NRPPa) may be used to carry the positioning information between NG-RAN and LMF **133** over a next-generation control plane interface (NG-C). In some aspects, LMF **133** configures the UE using the LTE positioning protocol (LPP) via AMF **132**. The NG-RAN configures the UE **101** using radio resource control (RRC) protocol over LTE-Uu and NR-Uu interfaces.

[0040] In some aspects, the 5G system architecture **140**B configures different reference signals to enable positioning measurements. Example reference signals that may be used for positioning measurements include the positioning reference signal (NR PRS) in the downlink and the sounding reference signal (SRS) for positioning in the uplink. The downlink positioning reference signal (PRS) is a reference signal configured to support downlink-based positioning methods.

[0041] In some aspects, the 5G system architecture **140**B includes an IP multimedia subsystem (IMS) **168**B as well as a plurality of IP multimedia core network subsystem entities, such as call session control functions (CSCFs). More specifically, the IMS **168**B includes a CSCF, which can act as a proxy CSCF (P-CSCF) **162**BE, a serving CSCF (S-CSCF) **164**B, an emergency CSCF (E-CSCF) (not illustrated in FIG. **1**B), or interrogating CSCF (I-CSCF) **166**B. The P-CSCF **162**B can be configured to be the first contact point for the UE **102** within the IMS **168**B. The S-CSCF **164**B can be configured to handle the session states in the network, and the E-CSCF can be configured to handle certain aspects of emergency sessions such as routing an emergency request to the correct emergency center or PSAP. The I-CSCF **166**B can be configured to function as the contact point within an operator's network for all IMS connections destined to a subscriber of that network operator, or a roaming subscriber currently located within that network operator's service area. In some aspects, the I-CSCF **166**B can be connected to another IP multimedia network **170**, e.g. an IMS operated by a different network operator.

[0042] In some aspects, the UDM/HSS **146** can be coupled to an application server (AS) **160**B, which can include a telephony application server (TAS) or another AS. The AS **160**B can be coupled to the IMS **168**B via the S-CSCF **164**B or the I-CSCF **166**B.

[0043] A reference point representation shows that interaction can exist between corresponding NF services. For example, FIG. **1**B illustrates the following reference points: N1 (between the UE **102** and the AMF **132**), N2 (between the RAN **110** and the AMF **132**), N3 (between the RAN **110** and the UPF **134**), N4 (between the SMF **136** and the UPF **134**), N5 (between the PCF **148** and the AF **150**, not shown), N6 (between the UPF **134** and the DN **152**), N7 (between the SMF **136** and the PCF **148**, not shown), N8 (between the UDM of UDM/HSS **146** and the AMF **132**, not shown), N9 (between two UPFs **134**, not shown), N10 (between the UDM of UDM/HSS **146** and the SMF **136**, not shown), N11 (between the AMF **132** and the SMF **136**, not shown), N12 (between the AUSF **144** and the AMF **132**, not shown), N13 (between the AUSF **144** and the UDM of UDM/HSS **146**, not shown), N14 (between two AMFs **132**, not shown), N15 (between the PCF **148** and the AMF **132** in case of a non-roaming scenario, or between the PCF **148** and a visited network and AMF **132** in case of a roaming scenario, not shown), N16 (between two SMFs, not shown), and N22 (between AMF **132** and NSSF **142**, not shown). Other reference point representations not shown in FIG. **1**B can also be used.

[0044] FIG. **1**C illustrates a 5G system architecture **140**C and a service-based representation. In addition to the network entities illustrated in FIG. **1**B, system architecture **140**C can also include a network exposure function (NEF) **154** and a network repository function (NRF) **156**. In some aspects, 5G system architectures can be service-based and interaction between network functions can be represented by corresponding point-to-point reference points Ni or as service-based interfaces.

[0045] In some aspects, as illustrated in FIG. **1**C, service-based representations can be used to represent network functions within the control plane that enable other authorized network functions to access their services. In this regard, 5G system architecture **140**C can include the following service-based interfaces: Namf **158**H (a service-based interface exhibited by the AMF **132**), Nsmf **158**I (a service-based interface exhibited by the SMF **136**), Nnef **158**B (a service-based interface exhibited by the NEF **154**), Npcf **158**D (a service-based interface exhibited by the PCF **148**), a Nudm **158**E (a service-based interface exhibited by the UDM of UDM/HSS **146**), Naf **158**F (a service-based interface exhibited by the AF **150**), Nnrf **158**C (a service-based interface exhibited by the NRF **156**), Nnssf **158**A (a service-based interface exhibited by the NSSF **142**), Nausf **158**G (a service-based interface exhibited by the AUSF **144**). Other service-based interfaces (e.g., Nudr, N5g-eir, and Nudsf) not shown in FIG. **1**C can also be used.

[0046] The figures illustrate various systems, devices, and components that may implement aspects of disclosed embodiments in different communication systems, such as 5G-NR networks including 5G non-terrestrial networks (NTNs). UEs, base stations (such as gNBs), and/or other nodes (e.g., satellites or other NTN nodes) discussed herein can be configured to perform the disclosed techniques.

[0047] FIG. **2** illustrates a network **200** in accordance with various embodiments. Network **200** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

[0048] The network **200** may include a UE **202**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **204** via an over-the-air connection. The UE **202** may be but is not limited to, a smartphone, tablet computer, wearable computing device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, a head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

[0049] In some embodiments, network **200** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

[0050] In some embodiments, the UE **202** may additionally communicate with an AP **206** via an over-the-air connection. The AP **206** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **204**. The connection between the UE **202** and the AP **206** may be consistent with any IEEE 802.11 protocol, wherein the AP **206** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **202**, RAN **204**, and AP **206** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **202** configured by the RAN **204** to utilize both cellular radio resources and WLAN resources.

[0051] The RAN **204** may include one or more access nodes, for example, access node (AN) **208**. AN **208** may terminate air-interface protocols for the UE **202** by providing access stratum protocols including RRC, Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), MAC, and L1 protocols. In this manner, the AN **208** may enable data/voice connectivity between the core network (CN) **220** and the UE **202**. In some embodiments, the AN **208** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **208** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **208** may be a macrocell base station or a low-power base station for providing femtocells, picocells, or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

[0052] In embodiments in which the RAN **204** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **204** is an LTE RAN) or an Xn interface (if the RAN **204** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

[0053] The ANs of the RAN **204** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **202** with an air interface for network access. The UE **202** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **204**. For example, the UE **202** and RAN **204** may use carrier aggregation to allow the UE **202** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG, and a second AN may be a secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

[0054] The RAN **204** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/SCells. Before accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

[0055] In V2X scenarios, the UE **202** or AN **208** may be or act as a roadside unit (RSU), which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by a UE may be referred to as a "UE-type RSU"; an eNB may be referred to as an "eNB-type RSU"; a gNB may be referred to as a "gNB-type RSU"; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, and media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high-speed events, such as crash avoidance, traffic warnings, and the like. Additionally, or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

[0056] In some embodiments, the RAN **204** may be an LTE RAN **210** with eNBs, for example, eNB **212**. The LTE RAN **210** may provide an LTE air interface with the following characteristics: sub-carrier spacing (SCS) of 15 kHz; CP-OFDM waveform for downlink (DL) and SC-FDMA waveform for uplink (UL); turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operate on sub-6 GHz bands.

[0057] In some embodiments, the RAN **204** may be an NG-RAN **214** with gNBs, for example, gNB **216**, or ng-eNBs, for example, ng-eNB **218**. The gNB **216** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **216** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **218** may also connect with the 5G core through an NG interface but may connect with a UE via an LTE air interface. The gNB **216** and the ng-eNB **218** may connect over an Xn interface.

[0058] In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **214** and a UPF **248** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **214** and an AMF **244** (e.g., N2 interface).

[0059] The NG-RAN **214** may provide a 5G-NR air interface with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM, and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH and tracking reference signal for time tracking. The 5G-NR air interface may operate on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include a synchronization signal and physical broadcast channel (SS/PBCH) block (SSB) which is an area of a downlink resource grid that includes PSS/SSS/PBCH.

[0060] In some embodiments, the 5G-NR air interface may utilize BWPs (bandwidth parts) for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **202** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **202**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **202** with different amounts of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with a small traffic load while allowing power saving at the UE **202** and in some cases at the gNB **216**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic loads.

[0061] The RAN **204** is communicatively coupled to CN **220** which includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **202**). The components of the CN **220** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **220** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **220** may be referred to as a network slice, and a logical instantiation of a portion of the CN **220** may be referred to as a network sub-slice.

[0062] In some embodiments, the CN **220** may be connected to the LTE radio network as part of the Enhanced Packet System (EPS) **222**, which may also be referred to as an EPC (or enhanced packet core). The EPC **222** may include MME **224**, SGW **226**, SGSN **228**, HSS **230**, PGW **232**, and PCRF **234** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the EPC **222** may be briefly introduced as follows.

[0063] The MME **224** may implement mobility management functions to track the current location of the UE **202** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

[0064] The SGW **226** may terminate an S1 interface toward the RAN and route data packets between the RAN and the EPC **222**. The SGW **226** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

[0065] The SGSN **228** may track the location of the UE **202** and perform security functions and access control. In addition, the SGSN **228** may perform inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **224**; MME selection for handovers; etc. The S3 reference point between the MME **224** and the SGSN **228** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

[0066] The HSS **230** may include a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **230** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **230** and the MME **224** may enable the transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **220**.

[0067] The PGW **232** may terminate an SGi interface toward a data network (DN) **236** that may include an application/content server **238**. The PGW **232** may route data packets between the LTE CN **220** and the data network **236**. The PGW **232** may be coupled with the SGW **226** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **232** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **232** and the data network **236** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for the provision of IMS services. The PGW **232** may be coupled with a PCRF **234** via a Gx reference point.

[0068] The PCRF **234** is the policy and charging control element of the LTE CN **220**. The PCRF **234** may be communicatively coupled to the app/content server **238** to determine appropriate QoS and charging parameters for service flows. The PCRF **234** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

[0069] In some embodiments, the CN **220** may be a 5GC **240**. The 5GC **240** may include an AUSF **242**, AMF **244**, SMF **246**, UPF **248**, NSSF **250**, NEF **252**, NRF **254**, PCF **256**, UDM **258**, and AF **260** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the 5GC **240** may be briefly introduced as follows.

[0070] The AUSF **242** may store data for the authentication of UE **202** and handle authentication-related functionality. The AUSF **242** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **240** over reference points as shown, the AUSF **242** may exhibit a Nausf service-based interface.

[0071] The AMF **244** may allow other functions of the 5GC **240** to communicate with the UE **202** and the RAN **204** and to subscribe to notifications about mobility events with respect to the UE **202**. The AMF **244** may be responsible for registration management (for example, for registering UE **202**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **244** may provide transport for SM messages between the UE **202** and the SMF **246**, and act as a transparent proxy for routing SM messages. AMF **244** may also provide transport for SMS messages between UE **202** and an SMSF. AMF **244** may interact with the AUSF **242** and the UE **202** to perform various security anchor and context management functions. Furthermore, AMF **244** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **204** and the AMF **244**; and the AMF **244** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **244** may also support NAS signaling with the UE **202** over an N3 IWF interface.

[0072] The SMF **246** may be responsible for SM (for example, session establishment, tunnel management between UPF **248** and AN **208**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **248** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **244** over N2 to AN **208**; and determining SSC mode of a session. SM may refer to the management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **202** and the data network **236**.

[0073] The UPF **248** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnecting to data network **236**, and a branching point to support multi-homed PDU sessions. The UPF **248** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **248** may include an uplink classifier to support routing traffic flows to a data network.

[0074] The NSSF **250** may select a set of network slice instances serving the UE **202**. The NSSF **250** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs if needed. The NSSF **250** may also determine the AMF set to be used to serve the UE **202**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **254**. The selection of a set of network slice instances for the UE **202** may be triggered by the AMF **244** with which the UE **202** is registered by interacting with the NSSF **250**, which may lead to a change of AMF. The NSSF **250** may interact with the AMF **244** via an N22 reference point and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **250** may exhibit an Nnssf service-based interface.

[0075] The NEF **252** may securely expose services and capabilities provided by 3GPP network functions for the third party, internal exposure/re-exposure, AFs (e.g., AF **260**), edge computing or fog computing systems, etc. In such embodiments, the NEF **252** may authenticate, authorize, or throttle the AFs. NEF **252** may also translate information exchanged with the AF **260** and information exchanged with internal network functions. For example, the NEF **252** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **252** may also receive information from other NFs based on the exposed capabilities of other NFs. This information may be stored at the NEF **252** as structured data, or a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **252** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **252** may exhibit a Nnef service-based interface.

[0076] The NRF **254** may support service discovery functions, receive NF discovery requests from NF instances, and provide information on the discovered NF instances to the NF instances. NRF **254** also maintains information on available NF instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during the execution of program code. Additionally, the NRF **254** may exhibit the Nnrf service-based interface.

[0077] The PCF **256** may provide policy rules to control plane functions to enforce them, and may also support a unified policy framework to govern network behavior. The PCF **256** may also implement a front end to access subscription information relevant to policy decisions in a UDR of the UDM **258**. In addition to communicating with functions over reference points as shown, the PCF **256** exhibits an Npcf service-based interface.

[0078] The UDM **258** may handle subscription-related information to support the network entities' handling of communication sessions and may store the subscription data of UE **202**. For example, subscription data may be communicated via an N8 reference point between the UDM **258** and the AMF **244**. The UDM **258** may include two parts, an application front end, and a UDR. The UDR may store subscription data and policy data for the UDM **258** and the PCF **256**, and/or structured data for exposure and application data (including PFDs for application detection, and application request information for multiple UEs) for the NEF **252**. The Nudr service-based interface may be exhibited by the UDR to allow the UDM **258**, PCF **256**, and NEF **252** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to the notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management, and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **258** may exhibit the Nudm service-based interface.

[0079] The AF **260** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

[0080] In some embodiments, the 5GC **240** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **202** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **240** may select a UPF **248** close to the UE **202** and execute traffic steering from the UPF **248** to data network **236** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **260**. In this way, the AF **260** may influence UPF (re) selection and traffic routing. Based on operator deployment, when AF **260** is considered to be a trusted entity, the network operator may permit AF **260** to interact directly with relevant NFs. Additionally, the AF **260** may exhibit a Naf service-based interface.

[0081] The data network **236** may represent various network operator services, Internet access, or third-party services that may be provided by one or more servers including, for example, application/content server **238**.

[0082] In some aspects, network **200** is configured for NR positioning using the location management function (LMF) **245**, which can be configured as an LMF node or as functionality in a different type of node. In some embodiments, LMF **245** is configured to receive measurements and assistance information from NG-RAN **214** and UE **202** via the AMF **244** (e.g., using an NLs interface) to compute the position of the UE. In some embodiments, NR positioning protocol A (NRPPa) protocol can be used for carrying the positioning information between NG-RAN **214** and LMF **245** over a next-generation control plane interface (NG-C).

In some embodiments, LMF **245** configures the UE **202** using LTE positioning protocol (LPP) (e.g., LPP-based communication link) via the AMF **244**. In some aspects, NG-RAN **214** configures the UE **202** using, e.g., radio resource control (RRC) protocol signaling over, e.g., LTE-Uu and NR-Uu interfaces. In some aspects, UE **202** uses the LTE-Uu interface to communicate with the ng-eNB **218** and the NR-Uu interface to communicate with the gNB **216**. In some aspects, ng-eNB **216** and gNB **216** use NG-C interfaces to communicate with the AMF **244**.

[0083] In some embodiments, the following reference signals can be used to achieve positioning measurements in NR communication networks: NR positioning reference signal (NR PRS) in the downlink and sounding reference signal (SRS) for positioning in the uplink. The downlink positioning reference signal (PRS) can be used as a reference signal supporting downlink-based positioning techniques. In some aspects, the entire NR bandwidth can be covered by transmitting PRS over multiple symbols that can be aggregated to accumulate power.

[0084] FIG. **3** schematically illustrates a wireless network **300** in accordance with various embodiments. The wireless network **300** may include a UE **302** in wireless communication with AN **304**. The UE **302** and AN **304** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

[0085] The UE **302** may be communicatively coupled with the AN **304** via connection **306**. Connection **306** is illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHz frequencies.

[0086] The UE **302** may include a host platform **308** coupled with a modem platform **310**. The host platform **308** may include application processing circuitry **312**, which may be coupled with protocol processing circuitry **314** of the modem platform **310**. The application processing circuitry **312** may run various applications for the UE **302** that source/sink application data. The application processing circuitry **312** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example IP) operations.

[0087] The protocol processing circuitry **314** may implement one or more layer operations to facilitate the transmission or reception of data over connection **306**. The layer operations implemented by the protocol processing circuitry **314** may include, for example, MAC, RLC, PDCP, RRC, and NAS operations.

[0088] The modem platform **310** may further include digital baseband circuitry **316** that may implement one or more layer operations that are "below" layer operations performed by the protocol processing circuitry **314** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

[0089] The modem platform **310** may further include transmit circuitry **318**, receive circuitry **320**, RF circuitry **322**, and RF front end (RFFE) **324**, which may include or connect to one or more antenna panels **326**. Briefly, the transmit circuitry **318** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **320** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **322** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **324** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **318** receive circuitry **320**, RF circuitry **322**, RFFE **324**, and antenna panels **326** (referred to generically as "transmit/receive components") may be specific to details of a specific implementation such as, for example, whether the communication is TDM or FDM, in mmWave or sub-6 GHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed of in the same or different chips/modules, etc.

[0090] In some embodiments, the protocol processing circuitry **314** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

[0091] A UE reception may be established by and via the antenna panels **326**, RFFE **324**, RF circuitry **322**, receive circuitry **320**, digital baseband circuitry **316**, and protocol processing circuitry **314**. In some embodiments, the antenna panels **326** may receive a transmission from the AN **304** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **326**.

[0092] A UE transmission may be established by and via the protocol processing circuitry **314**, digital baseband circuitry **316**, transmit circuitry **318**, RF circuitry **322**, RFFE **324**, and antenna panels **326**. In some embodiments, the transmit components of the UE **302** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **326**.

[0093] Similar to the UE **302**, the AN **304** may include a host platform **328** coupled with a modem platform **330**. The host platform **328** may include application processing circuitry **332** coupled with protocol processing circuitry **334** of the modem platform **330**. The modem platform may further include digital baseband circuitry **336**, transmit circuitry **338**, receive circuitry **340**, RF circuitry **342**, RFFE circuitry **344**, and antenna panels **346**. The components of the AN **304** may be similar to and substantially interchangeable with the like-named components of the UE **302**. In addition to performing data transmission/reception as described above, the components of the AN **304** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

[0094] FIG. **4** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **4** shows a diagrammatic representation of hardware resources **400** including one or more processors (or processor cores) **410**, one or more memory/storage devices **420**, and one or more communication resources **430**, each of which may be communicatively coupled via a bus **440** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **402** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **400**.

[0095] The one or more processors **410** may include, for example, a processor **412** and a processor **414**. The one or more processors **410** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

[0096] The memory/storage devices **420** may include a main memory, disk storage, or any suitable combination thereof. The memory/storage devices **420** may include but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

[0097] The communication resources **430** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **404** or one or more databases **406** or other network elements via a network **408**. For example, the communication resources **430** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

[0098] Instructions **450** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the one or more processors **410** to perform any one or more of the methodologies discussed herein. The instructions **450** may reside, completely or partially, within at least one of the one or more processors **410** (e.g., within the processor's cache memory), the memory/storage devices **420**, or any suitable combination thereof. Furthermore, any portion of the instructions **450** may be transferred to the hardware resources **400** from any combination of the peripheral devices **404** or the databases **406**. Accordingly, the memory of the one or more processors **410**, the memory/storage devices **420**, the peripheral devices **404**, and the databases **406** are examples of computer-readable and machine-readable media.

[0099] For one or more embodiments, at least one of the components outlined in one or more of the figures may be configured to perform one or more operations, techniques, processes, and/or methods as outlined in the example sections below. For example, baseband circuitry associated with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, satellite, network element, etc. as described in connection with one or more of the figures may be configured to operate in accordance with the disclosed techniques including one or more of the examples set forth below in the example section.

[0100] The term "application" may refer to a complete and deployable package, or environment to achieve a certain function in an operational environment. The term "AI/ML application" or the like may be an application that contains some artificial intelligence (AI)/machine learning (ML) models and application-level descriptions. In some embodiments, an AI/ML application may be used for configuring or implementing one or more of the disclosed aspects.

[0101] The term "machine learning" or "ML" refers to the use of computer systems implementing algorithms and/or statistical models to perform a specific task(s) without using explicit instructions but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as "ML models" or the like) based on sample data (referred to as "training data," "model training information," or the like) to make predictions or decisions without being explicitly programmed to perform such tasks. Generally, an ML algorithm is a computer program that learns from experience concerning some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term "ML algorithm" refers to different concepts than the term "ML model," these terms as discussed herein may be used interchangeably for the present disclosure.

[0102] The term "machine learning model," "ML model," or the like may also refer to ML methods and concepts used by an ML-assisted solution. An "ML-assisted solution" is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), decision tree algorithms, support machine vectors, Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principal component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference. An "ML pipeline" is a set of functionalities, functions, or functional entities specific to an ML-assisted solution; an ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The "actor" is an entity that hosts an ML-assisted solution using the output of the ML model inference). The term "ML training host" refers to an entity, such as a network function, that hosts the training of the model. The term "ML inference host" refers to an entity, such as a network function, that hosts the model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML host informs the actor about the output of the ML algorithm, and the actor decides on an action (an "action" is performed by an actor as a result of the output of an ML-assisted solution). The term "model inference information" refers to information used as an input to the ML model for determining inference(s); the data used to train an

ML model and the data used to determine inferences may overlap, however, "training data" and "inference data" refer to different concepts.

[0103] Mobile communication has evolved significantly from early voice systems to today's highly sophisticated integrated communication platform. The next generation wireless communication systems, (e.g., 5G NR systems) will provide access to information and sharing of data anywhere, anytime by various users and applications. 5G NR is expected to be a unified network/system that targets to meet vastly different and sometimes conflicting performance dimensions and services. Such diverse multi-dimensional requirements are driven by different services and applications. In general, 5G NR will evolve based on 3GPP LTE-Advanced with additional potential new Radio Access Technologies (RATs) to enrich people's lives with better, simple, and seamless wireless connectivity solutions. NR will enable everything connected by wireless and deliver fast, rich content and services.

[0104] 5G NR supports highly precise positioning in the vertical and horizontal dimensions, which relies on timing-based, angle-based, power-based, or hybrid techniques to estimate the user location in the network. In particular, the following RAT-dependent positioning techniques can be used to meet the positioning requirements for various use cases, e.g., indoor, outdoor, and Industrial Internet-of-Thing (IoT):

[0105] (a) Downlink time difference of arrival (DL-TDOA);

[0106] (b) Uplink time difference of arrival (UL-TDOA);

[0107] (c) Downlink angle of departure (DL-AoD);

[0108] (d) Uplink angle of arrival (UL AoA);

[0109] (e) Multi-cell round trip time (multi-RTT); and

[0110] (f) NR enhanced cell ID (E-CID).

[0111] With wide bandwidth for positioning signal and beamforming capability in the mmWave frequency band, higher positioning accuracy can be achieved by RAT-dependent positioning techniques. In Rel-16, downlink positioning reference signal (DL-PRS) and uplink sounding reference signal (UL-SRS) for positioning were introduced as enablers to achieve target performance characteristics.

[0112] In Rel-18, to address use cases such as autonomous driving, sidelink, or vehicle-to-everything (V2X) based positioning is considered. More specifically, various scenarios including in-coverage, partial coverage, and out-of-network coverage can be considered for sidelink positioning. In some aspects, to meet the positioning accuracy requirement, a new sidelink reference signal, i.e., sidelink position reference signal (SL PRS), can be introduced.

[0113] FIG. **5** illustrates diagram **500** of an example of sidelink positioning with anchor UEs and a target UE, in accordance with some aspects. In the example of FIG. **5**, a target UE indicates the UE to be positioned while anchor UEs indicate the UEs supporting positioning of the target UE (e.g., by transmitting and/or receiving SL PRS and providing positioning-related information such as measurements based on the SL PRS). In some aspects, the SL PRS can be transmitted between anchor and target UEs for sidelink positioning.

[0114] In Rel-16, two modes of resource allocation can be used for sidelink communication that are applied in V2X applications: (1) mode 1 where the gNB allocates the resources for sidelink communications; and (2) mode 2 where the UEs autonomously select the resources for sidelink communication based on sensing and resource selection procedure.

[0115] In Rel-16, the resource pool can be defined for sidelink communication, where multiple UEs share the same resource pool for sidelink channel/signal transmission. For mode 2 resource allocation, the UEs autonomously select resources within the resource pool for physical sidelink control channel (PSCCH) and physical sidelink shared channel (PSSCH) transmission. In mode 1 resource allocation, the base station controls the resource allocation to the UEs. For sidelink positioning, a dedicated resource pool can be configured for SL PRS transmission. In this case, certain mechanisms can be defined on the configuration of the dedicated resource pool for SL PRS transmission.

[0116] For sidelink positioning, a similar mechanism can be considered for the transmission of SL PRS. In particular, SL PRS resource allocation may be controlled by the network or autonomously performed by the UEs that are involved in the sidelink positioning operation. In this case, certain mechanisms can be defined for resource allocation mode 1 and mode 2 for SL PRS transmission.

[0117] The disclosed techniques include mechanisms on the mode 1 resource allocation for sidelink positioning reference signals.

Mode 1 Resource Allocation for SL PRS Transmissions

[0118] For sidelink positioning, a mechanism based on the disclosed techniques can be considered for the transmission of SL PRS. In particular, SL PRS resource allocation may be controlled by the network or autonomously performed by the UEs that are involved in the sidelink positioning operation. In this case, certain mechanisms may be defined for resource allocation mode 1 and mode 2 for SL PRS transmission respectively.

[0119] Embodiments of resource allocation Mode 1 for SL PRS transmission are provided as described below. In the following embodiments, the SL PRS resource pool may refer to a dedicated resource pool for SL PRS.

[0120] In some aspects, for mode 1 where gNB allocates the resource for SL PRS transmission, a new DCI format may be defined to allocate/reserve the resource for SL PRS transmission and/or measurement report. In some aspects, for this option, the new DCI format may only allocate or reserve the resource for SL PRS transmission, i.e., no resource is allocated for PSSCH transmission.

[0121] In some embodiments, the new DCI format may allocate resources for SL PRS transmission and any associated PSCCH transmission. In a further example, the SL PRS and the associated PSCCH may be mapped to the same resource pool. Alternatively, the SL PRS and associated PSCCH may be mapped to different resource pools that may be configured by higher-layer signaling.

[0122] In the new DCI format, at least one or more following fields may be included: SL PRS resource pool index, time gap, SL PRS resource index for the initial transmission, SL PRS resource assignment, time resource assignment for SL PRS transmission, a number of repetitions for SL PRS transmission in an SL PRS resource pool, a repetition counter for SL PRS transmission, a QCL assumption for the transmission of SL PRS, a configuration index, and resource allocation for PSCCH (including PSCCH resource pool index if PSCCH and SL PRS are mapped to different resource pools and multiple resource pools for PSCCH transmission associated with SL PRS may be configured by higher layers).

[0123] In some aspects, for the SL PRS resource pool index, multiple SL PRS resource pools may be configured for SL positioning. The size of the SL PRS resource pool index can be $\lceil \log_2 I_{SL\text{-}Pos}, \rceil$, where $I_{SL\text{-}Pos}$ is the number of dedicated resource pools for SL PRS.

[0124] In some aspects, the time gap field can be used to determine the slot used for the PSCCH and initial SL PRS transmission. In particular, the time gap value $K_{SL\text{-}Pos}$ can be determined based on the indication in a (pre-) configured time gap table for SL PRS transmission. Further, the slot for the PSCCH and initial SL PRS transmission can be the first slot of the corresponding dedicated resource pool for SL PRS that starts not earlier than

$$T_{DL} - \frac{T_{TA}}{2} + K_{SL-Pos} \times T_{slot},$$

where T_{DL} is the starting time of the downlink slot carrying the corresponding DCI, T_{TA} is the timing advance value corresponding to the TAG of the serving cell on which the DCI is received and $K_{SL\text{-}Pos}$ is the slot offset between the slot of the DCI and the first sidelink transmission scheduled by DCI and T_{slot} is the SL slot duration.

[0125] FIG. **6** illustrates diagram **600** of a time gap between a downlink control information (DCI) format, a sidelink control information (SCI) format, and a sidelink positioning reference signal (SL PRS) transmission, in accordance with some aspects. In the example of FIG. **6**, the 5th slot is configured for the SL PRS resource pool. Further, a new DCI format can be used to allocate/reserve the resource for SL PRS transmission. Based on the time gap indication in the DCI format, $K_{SL\text{-}Pos}=2$. In this case, SL PRS is transmitted in the 5th slot within the configured SL PRS resource pool.

[0126] In some aspects, the SL PRS resource index for initial transmission can be included in the DCI format. The size of the SL PRS resource index can be $\lceil \log_2 N_{SL\text{-}Pos}, \rceil$, where $N_{SL\text{-}Pos}$ is the number of SL PRS resources within an SL PRS resource pool.

[0127] In some aspects, SL PRS resource assignment can be used to determine the SL PRS resource index within an SL PRS resource pool for initial transmission and subsequent transmission in allocated/reserved resources. The equation to derive the SL PRS resource assignment can be based on the y:

- **[0128]** (a) If sl-MaxNumPerReserve is 2 then $PFIV=n_{SL\text{-}Pos,1}$; and
- **[0129]** (b) If sl-MaxNumPerReserve is 3 then $PFIV=n_{SL\text{-}Pos,1}+N_{SL\text{-}Pos}\cdot N_{SL\text{-}Pos,2}$, where $n_{SL\text{-}Pos,1}$ and $n_{SL\text{-}Pos,2}$ are the SL PRS resource index for the second and third time and resource or SL PRS resource pool; $N_{SL\text{-}Pos}$ is the number of SL PRS resources within an SL PRS resource pool.

[0130] In some aspects, the SL PRS resource index may be represented as starting comb offset for an SL PRS transmission in an SL PRS resource pool. In one option, when multiple frequency resources are configured within an SL PRS resource pool, where each frequency resource is used for an SL PRS transmission, the frequency resource index and starting comb offset may be separately allocated/reserved for SL PRS transmission. This can also apply to the case when multiple time domain resources are configured within an SL PRS resource pool.

[0131] In some aspects, a similar mechanism as defined for time resource assignment for SL communication can be reused for SL PRS transmission. In particular, depending on the number of reserved resources, the equation can be derived in accordance with Clause 8.1.5 in 3GPP TS 38.214. In some aspects, the slot index is determined based on the set of slots allocated for the SL PRS resource pool.

[0132] FIG. **7** illustrates a diagram **700** of resource allocation and reservation for a SL PRS transmission, in accordance with some aspects. In the example of FIG. **7**, the maximum number of reserved resources is 3 and the number of SL PRS resources in an SL PRS resource pool is 4 (index #0, #1, #2, and #3). Based on the indication in the DCI format, the UE can determine that the initial SL PRS transmission is in the 1st slot and SL PRS resource #1. Further, two reserved resources for SL PRS transmissions are in {5th slot, SL PRS resource #2} and {30th slot, SL PRS resource #0}. Note that in the figure, each time unit represents a slot.

[0133] In some aspects, several repetitions for SL PRS transmission in an SL PRS resource pool can be included in the DCI format for scheduling SL PRS transmission. Alternatively, the number of repetitions for an SL PRS transmission can be part of the SL PRS resource or SL PRS resource pool configuration, which can be configured by higher layers. When an anchor UE transmits the SL PRS in an SL PRS resource, the UE may transmit the SL PRS with and without repetitions in accordance with the SL PRS resource configuration.

[0134] In some aspects, SL PRS with or without any associated PSCCH transmission may be scheduled with repetitions following a repetition pattern with a period of SL PRS resources, wherein the period of multiple SL PRS resources, denoted by $T_{per}^{SL\text{-}PRS}$, may be provided by higher layers, a number of repetitions, $T_{rep}^{SL\text{-}PRS}$, and resource gaps, $T_{gap}^{SL\text{-}PRS}$, provided by higher layers. In another variation, the values of one or more of: $T_{per}^{SL\text{-}PRS}$, $T_{rep}^{SL\text{-}PRS}$, and $T_{gap}^{SL\text{-}PRS}$ may be provided by a combination of higher-layer signaling and dynamic signaling. In some aspects, such signaling may be supported not only for the case of a new DCI format used to schedule SL PRS transmissions but also when DCI format 3_0 is used with CRC scrambled with an RNTI that is the same as or different from that for SL communication resource scheduling.

[0135] In some aspects, the Quasi Co-Location (QCL) assumption for the transmission of SL PRS can be included in the DCI format for scheduling SL PRS transmission, which defines quasi-co-location information of the SL PRS resource with other reference signals. The SL PRS may be configured with QCL 'typeD' with another SL PRS resource, or with rs-Type set to 'typeC', 'typeD', or 'typeC-plus-typeD' with an SL SS/PBCH Block or DMRS associated with PSCCH or PSSCH transmission from a UE. Alternatively, QCL assumption for transmission of SL PRS may be included via higher layer signaling, e.g., as part of the configuration of the SL PRS resource pool.

[0136] In some aspects, a configuration index can be included in the DCI format for scheduling SL PRS transmission. In some aspects, more than one SL PRS configured grant type 2 configuration may be configured for SL PRS resource allocation. In this case, the configuration index in the DCI format may be used to index the SL PRS configured grant type 2.

[0137] In some aspects, if the UE is not configured to monitor the DCI format scheduling SL PRS with CRC scrambled by SL-CS-RNTI or SL-P-CS-RNTI, the configuration index may not be included or equal to 0 bit in the DCI format. If the UE is configured to monitor the DCI format with CRC scrambled by SL-CS-RNTI or SL-P-CS-RNTI, this field may be reserved for the DCI format with CRC scrambled by SL-RNTI or SL-P-RNTI.

[0138] In some aspects, a repetition counter can be included in the DCI format for scheduling SL PRS transmission. This indicates the repetition index for SL PRS transmission in different SL PRS resource pools. In one example, for initial SL PRS transmission, the repetition counter may be indicated as '0'. For the first SL PRS repetition, the repetition counter may be indicated as '1'; for the 2nd SL PRS repetition, the repetition counter may be indicated as '2', and so on.

[0139] In some aspects, resource allocation for PSCCH may include the sub-channel index for the initial transmission of PSCCH and time/frequency resource assignment for subsequent PSCCH transmissions. As a further extension, based on the association between the PSCCH resource index, e.g., sub-channel index and SL PRS resource index, the UE can determine the resource for initial and subsequent SL PRS transmissions.

[0140] In some embodiments, an SL-P-RNTI can be configured by higher layers for a UE to monitor the DCI format for scheduling SL PRS transmission, i.e., the CRC of the DCI format for scheduling SL PRS transmission is scrambled by an SL-P-RNTI, which is different from SL-RNTI or SL-CS-RNTI. In some aspects, the DCI format can be based on DCI format 3_0 for scheduling SL PRS transmission or a new DCI format as mentioned above.

[0141] In case when DCI format 3_0 is used to schedule SL PRS transmission, the DCI fields are only used for scheduling SL PRS transmission. The fields for scheduling PSCCH and PSSCH in one cell are not included. In this case, the DCI format may include one or more of the following fields: SL PRS resource pool index, a time gap, SL PRS resource index for the initial transmission, SL PRS resource assignment, time resource assignment for SL PRS transmission, a number of repetitions for SL PRS transmission in an SL PRS resource pool, a repetition counter for SL PRS transmission, QCL assumption for the transmission of SL PRS, and a configuration index.

[0142] In some aspects, DCI format 3_0 may be used to schedule SL PRS transmission and any associated PSCCH transmission. In a further example, the SL PRS and the associated PSCCH may be mapped to the same resource pool. Alternatively, the SL PRS and associated PSCCH may be mapped to different resource pools that may be provided by higher layers. In such a case, the DCI format may also include resource allocation information for PSCCH transmission in addition to the one or more fields listed above.

[0143] Further, if the total payload size of the DCI fields for scheduling SL PRS transmission is less than that for scheduling PSCCH and PSSCH, reserved fields or zero padding bits may be included in the DCI format to match the payload size for scheduling PSCCH and PSSCH.

[0144] Similarly, if the total payload size of the DCI fields for scheduling PSCCH and PSCCH is less than that for scheduling SL PRS transmission, padding bits may be included in the DCI format to match the payload size for scheduling SL PRS.

[0145] In some aspects, when a DCI format for scheduling SL PRS transmission is used for resource allocation and reservation of SL PRS resource, and if more than one SL PRS resource pools are (pre-) configured by higher layers, zeros shall be appended to the DCI format until the payload size is equal to the size of a DCI format given by a configuration of the SL PRS resource pool resulting in the largest number of information bits for the DCI format. In some embodiments, the DCI format can be based on DCI format 3_0 for scheduling SL PRS transmission or a new DCI format as mentioned above.

[0146] In some aspects, when the UE is configured to monitor the new DCI format for scheduling SL PRS transmission and DCI format 3_0 and/or DCI format 3_1, DCI sizes for the DCI formats are aligned.

[0147] In one option, if the UE is configured to monitor one of the DCI formats 3_0 and 3_1, and the new DCI format for scheduling SL PRS transmission, if the number of information bits in DCI format 3_0 or 3_1 is less than the payload of the new DCI format, zeros shall be appended to DCI format 3_0 or 3_1 until the payload size equals that of the new DCI format. Similarly, if the number of information bits in the new DCI format is less than the payload of the DCI format 3_0 or 3_1, zeros shall be appended to the new DCI format for scheduling SL PRS transmission until the payload size equals that of the DCI format 3_0 or 3_1.

[0148] In another option, if the UE is configured to monitor DCI format 3_0, DCI format 3_1, and the new DCI format for scheduling SL PRS transmission, if the number of information bits in one of the DCI formats is less than the maximum payload of the DCI format 3_0, DCI format 3_1, and the new DCI format, zeros shall be appended to one of the DCI formats until the payload size equals that of the maximum payload size.

[0149] In some aspects, for mode 1 where gNB allocates the resource for SL PRS transmission, DCI format 3_0 may be extended to support the SL PRS resource allocation.

[0150] In some aspects, one field in the DCI format 3_0 can be included to indicate the triggering of SL PRS transmission and/or measurement report. In particular, bit "1" may be used to indicate that the SL PRS transmission and/or measurement report is triggered, while bit "0" may be used to indicate that the SL PRS transmission and/or measurement report is not triggered.

[0151] In the following, a "dedicated resource pool" is used to refer to an SL resource pool that is dedicated to the transmission and/or reception of SL PRS and any SL channels/signals associated with the SL PRS. On the other hand, a "shared resource pool" is used to refer to an SL resource pool that may be used for SL communications and/or for transmission and/or reception of SL PRS and any SL channels/signals.

[0152] For this option, if a shared resource pool is allocated for SL communication and SL PRS transmission, the same resource pool index and time gap value can be used to determine the resource pool for SL PRS and SL communication, and the resource for initial SL PRS transmission, respectively. Further, frequency resource assignment and time resource assignment can be used to indicate the allocated and reserved resources for SL communication and SL PRS transmission. Here, "SL communication" may correspond to only PSSCH or both PSCCH and PSSCH. For the first option, separate time and frequency domain resources may be indicated for PSCCH (similar to Rel-16 specifications for PSCCH with stage-1 SCI), distinct from the resources for PSSCH.

[0153] If a dedicated resource pool is (pre-) configured for SL PRS transmission, the time gap between DCI format or PSCCH/PSSCH transmission and SL PRS transmission may be (pre-) configured by higher layers or dynamically indicated in the DCI or a combination thereof. Alternatively, the time gap can be determined per predefined rules.

[0154] In one option, the first slot for SL PRS transmission can be the first slot allocated for the SL PRS resource pool that is K slots after the first slot for PSCCH/PSSCH transmission, where slot duration for the K slots can be determined in accordance with the subcarrier spacing configured for SL PRS resource pool, SL communication resource pool, or SL BWP for SL PRS and/or SL communication. In one example, the first slot for SL PRS transmission is the first slot for the SL PRS resource pool after the first slot for PSCCH/PSSCH transmission, i.e., K=0.

[0155] FIG. **8** illustrates a diagram **800** of a time gap between a DCI format, a PSCCH/PSSCH transmission, and an SL PRS transmission, in accordance with some aspects. In the example of FIG. **8**, the 4th and 7th slots are allocated for the SL communication resource pool and SL PRS resource pool, respectively. Further, an extension of DCI format 3_0 can be used for PSCCH/PSSCH and SL PRS resource allocation. K=1 can be configured by the higher layers for the time gap between resources for SL communication and SL PRS transmission. In this case, SL PRS can be transmitted in the 7th slot in accordance with the aforementioned rule.

[0156] In some aspects, one field in the DCI format 3_0 can be included to indicate the resource for the initial SL PRS transmission within an SL PRS resource pool. In some aspects, the same or different SL PRS resource index may be applied for the initial SL PRS transmission or subsequent SL PRS transmission in the reserved resources.

[0157] In some aspects, time resource assignment and SL PRS resource assignment as mentioned above may be included in the DCI format 3_0 to indicate the slot index and SL PRS resource index for the reserved resource for SL PRS transmission.

[0158] In some embodiments, for configured grant type 2 of SL PRS resource allocation, a separate SL-P-CS-RNTI may be configured by higher layers for scheduling activation or scheduling release, where the CRC of a DCI format 3_0 or a new DCI for scheduling SL PRS transmission is scrambled with SL-P-CS-RNTI.

[0159] In some aspects, for activation of SL PRS configured grant Type 2, the validation of the DCI format 3_0 and/or the new DCI for scheduling SL PRS transmission is achieved if some fields are set to a special code point. In one example, the existing code point as defined in Clause 10.2A in 3GPP TS 38.214 can be reused for activation of SL PRS configured grant Type 2 when the DCI format is extended to allocate/reserve the resource for SL PRS transmission.

[0160] For the release of SL PRS configured grant Type 2, the validation of the DCI format 3_0 and/or the new DCI for scheduling SL PRS transmission is achieved if some fields are set to a special code point. In one example, the existing code point as defined in Clause 10.2A in TS 38.214 can be reused for the release of SL PRS configured grant Type 2 when the DCI format is extended to allocate/reserve the resource for SL PRS transmission.

[0161] In some aspects, SL PRS resource assignment can be set to all "1". In another example, the SL PRS resource index for the initial transmission can be set to all "0" or all "1".

[0162] In some aspects, DCI format 3_0 may be extended for SL PRS resource allocation in both the shared resource pool and the dedicated resource pool.

[0163] In some embodiments, a one-bit field may be included in the DCI format 3_0 to indicate whether SL PRS transmission is in a shared resource pool or a dedicated resource pool. In particular, bit "1" may be used to indicate that the DCI format 3_0 is for SL PRS resource allocation in a shared resource pool, while bit "0" may be used to indicate that the DCI format 3_0 is for SL PRS resource allocation in a dedicated resource pool. This option may apply when a single resource pool index may be used for a shared and dedicated resource pool.

[0164] In this case, when the one-bit indicator differentiates SL PRS resource allocation in a shared resource pool and dedicated resource pool, the one-bit indicator may be located at the beginning of DCI format 3_0. Further, when the DCI format 3_0 is for a dedicated resource pool in accordance with the one-bit indicator, the remaining fields are repurposed for SL PRS resource allocation in a dedicated resource pool. The above embodiments on the DCI fields for SL PRS resource allocation can apply.

[0165] In some aspects, for the case of dedicated resource pools, the fields related to resource allocation are reinterpreted to indicate the remaining information for SL PRS resources regarding the time, frequency domain, and comb-offset resource allocation.

[0166] In some embodiments, a resource pool index is used to indicate either a shared resource pool or a dedicated resource pool. In an example of the embodiment, a first set of resource pool indices are configured for shared resource pools and a second set of resource pool indexes are configured for dedicated resource pool. In one example, resource pool indexes #0-5 may be configured for shared resource pools while resource pool indexes #6-7 may be configured for a dedicated resource pool.

[0167] In some aspects, a resource pool index may be associated with a specific type of SL PRS resource pool (i.e., shared or dedicated) via higher layer (pre)-configuration, and a single value of SL resource pool index may map to a given resource pool. In this case, the resource pool indices may not be partitioned via specifications to correspond to one of a dedicated or shared resource pool(s).

[0168] In this case, when the resource pool index is pointed to a dedicated resource pool, the remaining fields after the resource pool index bit-field may be repurposed for SL PRS resource allocation in a dedicated resource pool. The above embodiments on the DCI fields for scheduling SL PRS transmission can apply. In addition, a single DCI format size is maintained regardless of whether the DCI format 3_0 is used for resource allocation in a dedicated resource pool or a shared resource pool. When the DCI format size for DCI format 3_0 for a dedicated resource pool is less than that for a shared resource pool, zero padding is applied to the DCI format for the dedicated resource pool to match the DCI format size for the shared resource pool, and vice versa.

[0169] In some aspects, the SL PRS resource configuration may be different from the shared resource pool and dedicated resource pool.

[0170] In some aspects, DCI format 3_0 may be extended for SL PRS resource allocation in a shared resource pool, while a new DCI format may be defined for SL PRS resource allocation in a dedicated resource pool.

[0171] In some aspects, a one-bit field may be included in the DCI format 3_0 and the new DCI format to indicate whether the DCI format is for a dedicated resource pool or a shared resource pool. In particular, bit "1" may be used to indicate that the DCI format is for SL PRS resource allocation in a shared resource pool, while bit "0" may be used to indicate that the DCI format is for SL PRS resource allocation in a dedicated resource pool. This option may apply when a single resource pool index may be used for a shared and dedicated resource pool.

[0172] In some embodiments, a resource pool index is used to indicate either a shared resource pool or a dedicated resource pool. In an example of the embodiment, a first set of resource pool indices are configured for shared resource pools and a second set of resource pool indexes are configured for dedicated resource pool. In one example, resource pool indexes #0-5 may be configured for shared resource pools while resource pool indexes #6-7 may be configured for a dedicated resource pool.

[0173] In some aspects, which resource pool index may be associated with a specific type of SL PRS resource pool (i.e., shared or dedicated) via higher layer (pre)-configuration and a single value of SL resource pool index may map to a given resource pool. In this case, the resource pool indices may not be partitioned via specifications to correspond to one of a dedicated or shared resource pool(s).

[0174] In some aspects, based on the resource pool index, UE may determine whether this is for DCI format 3_0 or the new DCI format for SL PRS resource allocation.

[0175] In some embodiments, whether to monitor DCI format 3_0 or the new DCI format may be configured by higher layers via RRC signaling. For instance, UE may only monitor either DCI format 3_0 for SL PRS resource allocation in a shared resource pool or the new DCI format for SL PRS resource allocation in a dedicated resource pool.

[0176] In another option, the UE may be configured with a separate search space set for monitoring DCI format 3_0 and the new DCI format for SL PRS resource allocation.

[0177] In some aspects, the UE may be provided with different RNTI for monitoring DCI format 3_0 and the new DCI format for SL PRS resource allocation.

[0178] The disclosed techniques can further include the following features as well as the features disclosed in the examples listed below.

[0179] A system and method of wireless communication for a 5G NR system include allocating, by a gNB, a resource for sidelink positioning reference signal (SL PRS) transmission via a downlink control information (DCI) format using a model 1 resource allocation scheme. In some aspects, for mode 1 where gNB allocates the resource for SL PRS transmission, a new DCI format may be defined to allocate/reserve the resource for SL PRS transmission and measurement report.

[0180] In some aspects, in the new DCI format, at least one or more following fields may be included: SL PRS resource pool index, time gap, SL PRS resource index for the initial transmission, SL PRS resource assignment, time resource assignment for SL PRS transmission, number of repetitions for SL PRS transmission in an SL PRS resource pool, repetition counter for SL PRS transmission, QCL assumption for the transmission of SL PRS, configuration index, and resource allocation for PSCCH.

[0181] In some aspects, for the SL PRS resource pool index, more than one SL PRS resource pool may be configured for SL positioning.

[0182] In some aspects, the time gap field can be used to determine the slot used for the PSCCH and initial SL PRS transmission, where the time gap value $K_{SL\text{-}Pos}$ can be determined based on the indication of a (pre-) configured time gap table for SL PRS transmission.

[0183] In some aspects, the SL PRS resource index for initial transmission can be included in the DCI format.

[0184] In some aspects, SL PRS resource assignment can be used to determine the SL PRS resource index within an SL PRS resource pool for initial transmission and subsequent transmission in allocated/reserved resources.

[0185] In some aspects, several repetitions for SL PRS transmission in an SL PRS resource pool can be included in the DCI format for scheduling SL PRS transmission.

[0186] In some aspects, the Quasi-Co-Location (QCL) assumption for the transmission of SL PRS can be included in the DCI format for scheduling SL PRS transmission, which defines quasi-co-location information of the SL PRS resource with other reference signals.

[0187] In some aspects, more than one SL PRS configured grant type 2 configuration may be configured for SL PRS resource allocation.

[0188] In some aspects, a repetition counter can be included in the DCI format for scheduling SL PRS transmission, which indicates a repetition index for SL PRS transmission in different SL PRS resource pools.

[0189] In some aspects, resource allocation for PSCCH may include the sub-channel index for the initial transmission of PSCCH and time/frequency resource assignment for subsequent PSCCH transmissions.

[0190] In some aspects, an SL-P-RNTI can be configured by higher layers for a UE to monitor the DCI format for scheduling SL PRS transmission, i.e., the CRC of the DCI format for scheduling SL PRS transmission is scrambled by an SL-P-RNTI.

[0191] In some aspects, when the existing DCI format 3_0 is used to schedule SL PRS transmission, the DCI fields are only used for scheduling SL PRS transmission.

[0192] In some aspects, if the total payload size of the DCI fields for scheduling SL PRS transmission is less than that for scheduling PSCCH and PSSCH, reserved fields or zero padding bits may be included in the DCI format to match the payload size for scheduling PSCCH and PSSCH.

[0193] In some aspects, if the total payload size of the DCI fields for scheduling PSCCH and PSCCH is less than that for scheduling SL PRS transmission, padding bits may be included in the DCI format to match the payload size for scheduling SL PRS.

[0194] In some aspects, when a DCI format for scheduling SL PRS transmission is used for resource allocation and reservation of SL PRS resource, and if more than one SL PRS resource pools are (pre-) configured by higher layers, zeros shall be appended to the DCI format until the payload size is equal to the size of a DCI format given by a configuration of the SL PRS resource pool resulting in the largest number of information bits for the DCI format

[0195] In some aspects, when UE is configured to monitor the new DCI format for scheduling SL PRS transmission and DCI format 3_0 and/or DCI format 3_1, the DCI size for the DCI formats is aligned.

[0196] In some aspects, for mode 1 where gNB allocates the resource for SL PRS transmission, existing DCI format 3_0 may be extended to support the SL PRS resource allocation.

[0197] In some aspects, one field in the DCI format 3_0 can be included to indicate the triggering of SL PRS transmission and/or measurement report.

[0198] In some aspects, if a shared resource pool is allocated for SL communication and SL PRS transmission, the same resource pool index and time gap value can be used to determine the resource pool for SL PRS and SL communication, and the resource for initial SL PRS transmission, respectively.

[0199] In some aspects, the first slot for SL PRS transmission can be the first slot allocated for the SL PRS resource pool that is K slots after the first slot for PSCCH/PSSCH transmission, where slot duration for the K slots can be determined in accordance with the subcarrier spacing configured for SL PRS resource pool, SL communication resource pool, or SL BWP for SL PRS and/or SL communication.

[0200] In some aspects, for configured grant type 2 of SL PRS resource allocation, a separate SL-P-CS-RNTI may be configured by higher layers for scheduling activation or scheduling release, where the CRC of a DCI format 3_0 or a new DCI for scheduling SL PRS transmission is scrambled with SL-P-CS-RNTI.

[0201] In some aspects, for activation of SL PRS configured grant Type 2, the validation of the DCI format 3_0 and/or the new DCI for scheduling SL PRS transmission is achieved if some fields are set to a special code point.

[0202] In some aspects, for the release of SL PRS configured grant Type 2, the validation of the DCI format 3_0 and/or the new DCI for scheduling SL PRS transmission is achieved if some fields are set to a special code point.

[0203] In some aspects, a one-bit field may be included in the DCI format 3_0 to indicate whether SL PRS transmission is in a shared resource pool or a dedicated resource pool.

[0204] In some aspects, when the one-bit indicator differentiates SL PRS resource allocation in a shared resource pool and dedicated resource pool, the one-bit indicator may be located at the beginning of DCI format 3_0.

[0205] In some aspects, a resource pool index is used to indicate either a shared resource pool or a dedicated resource pool.

[0206] In some aspects, a first set of resource pool indices are configured for shared resource pools and a second set of resource pool indexes are configured for a dedicated resource pool.

[0207] In some aspects, a resource pool index may be associated with a specific type of SL PRS resource pool (i.e., shared or dedicated) via higher layer (pre)-configuration, and a single value of SL resource pool index may map to a given resource pool.

[0208] In some aspects, when the resource pool index is pointed to a dedicated resource pool, the remaining fields after the resource pool index bit-field may be repurposed for SL PRS resource allocation in a dedicated resource pool.

[0209] In some aspects, DCI format 3_0 may be extended for SL PRS resource allocation in a shared resource pool, while a new DCI format may be defined for SL PRS resource allocation in a dedicated resource pool.

[0210] In some aspects, a one-bit field may be included in the DCI format 3_0 and the new DCI format to indicate whether the DCI format is for a dedicated resource pool or a shared resource pool.

[0211] In some aspects, a resource pool index is used to indicate either a shared resource pool or a dedicated resource pool.

[0212] In some aspects, whether to monitor DCI format 3_0 or the new DCI format may be configured by higher layers via RRC signaling.

[0213] In some aspects, the UE may be configured with a separate search space set for monitoring DCI format 3_0 and the new DCI format for SL PRS resource allocation.

[0214] In some aspects, the UE may be provided with different RNTI for monitoring DCI format 3_0 and the new DCI format for SL PRS resource allocation.

[0215] FIG. **9** illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node such as a base station), a network-controlled repeater (NCR), an access point (AP), a wireless station (STA), a mobile station (MS), or user equipment (UE), in accordance with some aspects and to perform one or more of the techniques disclosed herein. In alternative aspects, the communication device **900** may operate as a standalone device or may be connected (e.g., networked) to other communication devices.

[0216] Circuitry (e.g., processing circuitry) is a collection of circuits implemented in tangible entities of the device **900** that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership may be flexible over time. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, the hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a machine-readable medium physically modified (e.g., magnetically, electrically, moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation.

[0217] In connecting the physical components, the underlying electrical properties of a hardware constituent are changed, for example, from an insulator to a conductor or vice versa. The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, in an example, the machine-readable medium elements are part of the circuitry or are communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in the first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry at a different time. Additional examples of these components with respect to the device **900** follow.

[0218] In some aspects, the device **900** may operate as a standalone device or may be connected (e.g., networked) to other devices. In a networked deployment, the communication device **900** may operate in the capacity of a server communication device, a client communication device, or both in server-client network environments. In an example, the communication device **900** may act as a peer communication device in a peer-to-peer (P2P) (or other distributed) network environment. The communication device **900** may be a UE, eNB, PC, a tablet PC, STB, PDA, mobile telephone, smartphone, a web appliance, network router, a switch or bridge, or any communication device capable of executing instructions (sequential or otherwise) that specify actions to be taken by that communication device. Further, while only a single communication device is illustrated, the term "communication device" shall also be taken to include any collection of communication devices that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), and other computer cluster configurations.

[0219] Examples, as described herein, may include, or may operate on, logic or several components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations and may be configured or arranged in a certain manner. In an example, circuits may be arranged (e.g., internally or with respect to external entities such as other circuits) in a specified manner as a module. In an example, the whole or part of one or more computer systems (e.g., a standalone, client, or server computer system) or one or more hardware processors may be configured by firmware or software (e.g., instructions, an application portion, or an application) as a module that operates to perform specified operations. In an example, the software may reside on a communication device-readable medium. In an example, the software, when executed by the underlying hardware of the module, causes the hardware to perform the specified operations.

[0220] Accordingly, the term "module" is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which modules are temporarily configured, each of the modules needs not to be instantiated at any one moment in time. For example, where the modules comprise a general-purpose hardware processor configured using the software, the general-purpose hardware processor may be configured as respective different modules at different times. The software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time.

[0221] The communication device (e.g., UE) **900** may include a hardware processor **902** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **904**, a static memory **906**, and a storage device **916** (e.g., hard drive, tape drive, flash storage, or other block or storage devices), some or all of which may communicate with each other via an interlink (e.g., bus) **908**.

[0222] The communication device **900** may further include a display device **910**, an alphanumeric input device **912** (e.g., a keyboard), and a user interface (UI) navigation device **914** (e.g., a mouse). In an example, the display device **910**, input device **912**, and UI navigation device **914** may be a touchscreen display. The communication device **900** may additionally include a signal generation device **918** (e.g., a speaker), a network interface device **920**, and one or more sensors **921**, such as a global positioning system (GPS) sensor, compass, accelerometer, or another sensor. The communication device **900** may include an output controller **928**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.).

[0223] The storage device **916** may include a communication device-readable medium **922**, on which is stored one or more sets of data structures or instructions **924** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. In some aspects, registers of the processor **902**, the main memory **904**, the static memory **906**, and/or the storage device **916** may be, or include (completely or at least partially), the device-readable medium **922**, on which is stored the one or more sets of data structures or instructions **924**, embodying or utilized by any one or more of the techniques or functions described herein. In an example, one or any combination of the hardware processor **902**, the main memory **904**, the static memory **906**, or the storage device **916** may constitute the device-readable medium **922**.

[0224] As used herein, the term "device-readable medium" is interchangeable with "computer-readable medium" or "machine-readable medium". While the communication device-readable medium **922** is illustrated as a single medium, the term "communication device-readable medium" may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **924**. The term "communication device-readable medium" is inclusive of the terms "machine-readable medium" or "computer-readable medium", and may include any medium that is capable of storing, encoding, or carrying instructions (e.g., instructions **924**) for execution by the communication device **900** and that causes the communication device **900** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Non-limiting communication device-readable medium examples may include solid-state memories and optical and magnetic media. Specific examples of communication device-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; Random Access Memory (RAM); and CD-ROM and DVD-ROM disks. In some examples, communication device-readable media may include non-transitory communication device-readable media. In some examples, communication device-readable media may include communication device-readable media that is not a transitory propagating signal.

[0225] Instructions **924** may further be transmitted or received over a communications network **926** using a transmission medium via the network interface device **920** utilizing any one of several transfer protocols. In an example, the network interface device **920** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **926**. In an example, the network interface device **920** may include a plurality of antennas to wirelessly communicate using at least one of the single-input-multiple-output (SIMO), MIMO, or multiple-input-single-output (MISO) techniques. In some examples, the network interface device **920** may wirelessly communicate using Multiple User MIMO techniques.

[0226] The term "transmission medium" shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the communication device **900**, and includes digital or analog communications signals or another intangible medium to facilitate communication of such software. In this regard, a transmission medium in the context of this disclosure is a device-readable medium.

[0227] The terms "machine-readable medium," "computer-readable medium," and "device-readable medium" mean the same thing and may be used interchangeably in this disclosure. The terms are defined to include both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals.

[0228] Described implementations of the subject matter can include one or more features, alone or in combination as illustrated below by way of examples.

[0229] Example 1 is an apparatus for user equipment (UE) configured for operation in a Fifth Generation New Radio (5G NR) network, the apparatus comprising: processing circuitry, wherein to configure the UE for mode 1 resource allocation for sidelink (SL) positioning in the 5G NR network, the processing circuitry is to: decode radio resource control (RRC) signaling or LTE positioning protocol (LPP) received from a base station for in-coverage scenario or sidelink positioning protocol (SLPP) signaling from a second UE or pre-configuration for out-of-coverage scenario, the LPP, RRC or SLPP signaling or pre-configuration respectively including configuration information configuring a resource pool associated with sidelink positioning reference signal (SL PRS); decode downlink control information (DCI) received from the base station via a physical downlink control channel (PDCCH), the DCI indicating a resource pool index and/or presence of a SL PRS resource from the indicated resource pool; and encode a SL PRS for transmission to a second UE based on the SL PRS resource; and a memory coupled to the processing circuitry and configured to store the configuration information and the DCI.

[0230] In Example 2, the subject matter of Example 1 includes subject matter where the processing circuitry is to: encode an SL PRS for transmission to a second UE, the transmission of the SL PRS using a time-frequency resource from the resource pool indicated by the DCI.

[0231] In Example 3, the subject matter of Example 2 includes subject matter where the processing circuitry is to: decode the DCI to further determine a dedicated resource pool index for SL PRS.

[0232] In Example 4, the subject matter of Example 3 includes subject matter where the processing circuitry is to: decode the LPP, RRC, or SLPP signaling or pre-configuration to determine the resource pool and at least a second resource pool associated with the SL PRS transmission.

[0233] In Example 5, the subject matter of Example 4 includes subject matter where the processing circuitry is to: perform a selection of one of the resource pool or the second resource pool using the SL PRS resource pool index; and determine the time-frequency resource based on the selection.

[0234] In Example 6, the subject matter of Examples 1-5 includes subject matter where the processing circuitry is to: decode the DCI to further determine a number of repetitions for an SL PRS transmission in the resource pool.

[0235] In Example 7, the subject matter of Example 6 includes subject matter where the processing circuitry is to: encode a second SL PRS for multiple transmissions to the second UE, the multiple transmissions based on the number of repetitions configured by the DCI.

[0236] In Example 8, the subject matter of Examples 1-7 includes subject matter where the resource pool is a shared resource pool allocated for SL communication and transmission of an SL PRS.

[0237] In Example 9, the subject matter of Example 8 includes subject matter where the processing circuitry is to: decode the DCI to determine a resource pool index and a time gap value; select the resource pool as the shared resource pool based on the resource pool index; and determine a resource from the shared resource pool for an initial SL PRS transmission based at least on the time gap value.

[0238] In Example 10, the subject matter of Examples 1-9 includes, transceiver circuitry coupled to the processing circuitry; and at least one antenna coupled to the transceiver circuitry, wherein the processing circuitry causes transmission of the SL PRS using the at least one antenna.

[0239] Example 11 is a computer-readable storage medium that stores instructions for execution by one or more processors of a base station, the instructions to configure the base station for mode 1 resource allocation for sidelink (SL) positioning in a Fifth Generation New Radio (5G NR) network, and to cause the base station to perform operations comprising: encoding radio resource control (RRC) signaling for transmission to user equipment (UE), the RRC signaling including configuration information configuring a resource pool associated with sidelink positioning reference signal (SL PRS); and encoding downlink control information (DCI) for transmission to the UE using a physical downlink control channel (PDCCH), the DCI indicating a resource pool index and/or presence of an SL PRS from the indicated resource pool.

[0240] In Example 12, the subject matter of Example 11 includes, the operations further comprising: decoding a second SL-PRSSL PRS received from the UE, the second SL-PRSSL PRS received based on a second time-frequency resource from the resource pool indicated by the DCI.

[0241] Example 13 is a computer-readable storage medium that stores instructions for execution by one or more processors of user equipment (UE), the instructions to configure the UE for mode 1 resource allocation for sidelink (SL) positioning in a Fifth Generation New Radio (5G NR) network, and to cause the UE to perform operations comprising: decoding radio resource control (RRC) signaling or LTE positioning protocol (LPP) received from a base station for in-coverage scenario or sidelink positioning protocol (SLPP) signaling from a second UE or pre-configuration for out-of-coverage scenario, the LPP, RRC or SLPP signaling or pre-configuration respectively including configuration information configuring a resource pool associated with sidelink positioning reference signal (SL PRS); decoding downlink control information (DCI) received from the base station via a physical downlink control channel (PDCCH), the DCI indicating a resource pool index and/or presence of a SL PRS from the indicated resource pool; and encode a SL PRS transmission to a second UE based on the SL PRS resource.

[0242] In Example 14, the subject matter of Example 13 includes, the operations further comprising: encoding an SL PRS for transmission to the second UE, and the transmission of the SL PRS using a time-frequency resource from the resource pool indicated by the DCI.

[0243] In Example 15, the subject matter of Example 14 includes, the operations further comprising: decoding the DCI to further determine an SL PRS resource pool index.

[0244] In Example 16, the subject matter of Example 15 includes, the operations further comprising: decoding the RRC, LPP, SLPP signaling or pre-configuration to determine the resource pool and at least a second resource pool associated with the sidelink positioning reference signal (SL PRS); performing a selection of one of the resource pool or the second resource pool using the SL PRS resource pool index; and determining the time-frequency resource based on the selection.

[0245] In Example 17, the subject matter of Example 16 includes, the operations further comprising: performing a selection of one of the resource pool or the second resource pool using the SL PRS resource pool index; and determining the time-frequency resource based on the selection.

[0246] In Example 18, the subject matter of Examples 13-17 includes, the operations further comprising: decoding the DCI to further determine a number of repetitions for an SL PRS transmission in the resource pool.

[0247] In Example 19, the subject matter of Example 18 includes, the operations further comprising: encoding a second SL PRS for multiple transmissions to the second UE, the multiple transmissions based on the number of repetitions configured by the DCI.

[0248] In Example 20, the subject matter of Examples 13-19 includes subject matter where the resource pool is a shared resource pool allocated for SL communication and transmission of a second SL PRS, and the operations further comprising: decoding the DCI to determine a resource pool index and a time gap value; selecting the resource pool as the shared resource pool based on the resource pool index; and determining a resource from the shared resource pool for an initial SL PRS transmission based at least on the time gap value.

[0249] Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.

[0250] Example 22 is an apparatus comprising means to implement any of Examples 1-20.

[0251] Example 23 is a system to implement any of Examples 1-20.

[0252] Example 24 is a method to implement any of Examples 1-20.

[0253] Although an aspect has been described concerning specific exemplary aspects, it will be evident that various modifications and changes may be made to these aspects without departing from the broader scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various aspects is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

1-20. (canceled)

21. An apparatus for a user equipment (UE) configured for operation in a Fifth Generation New Radio (5G NR) network, the apparatus comprising:

processing circuitry, wherein to configure the UE for sidelink positioning in the 5G NR network, the processing circuitry is to:

decode higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools;

decode a downlink control information (DCI) format received from the base station, the DCI format to configure scheduling information of a physical sidelink control channel (PSCCH) and a SL PRS, the scheduling information based on a SL PRS resource pool of the plurality of SL PRS resource pools;

decode SL control data associated with the PSCCH and the SL PRS, the SL control data and the SL PRS received from a second UE using the SL PRS resource pool; and

encode a second SL PRS for an SL PRS transmission to the second UE using the SL PRS resource pool; and

a memory coupled to the processing circuitry and configured to store the DCI format.

22. The apparatus of claim **21**, wherein the DCI format is a DCI format 3_0.

23. The apparatus of claim **22**, wherein the SL PRS resource pool of the plurality of SL PRS resource pools indicated by the DCI format 3_0 is a shared SL PRS pool.

24. The apparatus of claim **23**, wherein the DCI format 3_0 further includes a resource pool index of the shared SL PRS pool.

25. The apparatus of claim **24**, wherein a size of the resource pool index is based on a number of the plurality of SL PRS resource pools.

26. The apparatus of claim **21**, wherein the SL PRS resource pool of the plurality of SL PRS resource pools is a dedicated SL PRS pool.

27. The apparatus of claim **21**, wherein the processing circuitry is to:

decode the DCI format to further determine a number of repetitions for the SL PRS transmission in the SL PRS resource pool.

28. The apparatus of claim **27**, wherein the processing circuitry is to:

decode the DCI format to determine a resource pool index and a time gap value;

select the SL PRS resource pool as a shared SL PRS resource pool based on the resource pool index; and

determine a resource from the shared SL PRS resource pool for the SL PRS transmission based at least on the time gap value.

29. The apparatus of claim **21**, further comprising:

transceiver circuitry coupled to the processing circuitry; and

one or more antennas coupled to the transceiver circuitry.

30. A non-transitory computer-readable storage medium that stores instructions for execution by one or more processors of a user equipment (UE), the instructions to configure the UE for sidelink positioning in a Fifth Generation New Radio (5G NR) network, and to cause the UE to perform operations comprising:

decoding higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools;

decoding a downlink control information (DCI) format received from the base station, the DCI format to configure scheduling information of a physical sidelink control channel (PSCCH) and a SL PRS, the scheduling information based on a SL PRS resource pool of the plurality of SL PRS resource pools;

decoding SL control data associated with the PSCCH and the SL PRS, the SL control data and the SL PRS received from a second UE using the SL PRS resource pool; and

encoding a second SL PRS for an SL PRS transmission to the second UE using the SL PRS resource pool; and

31. The non-transitory computer-readable storage medium of claim **30**, wherein the DCI format is a DCI format 3_0.

32. The non-transitory computer-readable storage medium of claim **31**, wherein the SL PRS resource pool of the plurality of SL PRS resource pools indicated by the DCI format 3_0 is a shared SL PRS pool.

33. The non-transitory computer-readable storage medium of claim **32**, wherein the DCI format 3_0 further includes a resource pool index of the shared SL PRS pool.

34. The non-transitory computer-readable storage medium of claim **33**, wherein a size of the resource pool index is based on a number of the plurality of SL PRS resource pools.

35. The non-transitory computer-readable storage medium of claim **30**, wherein the SL PRS resource pool of the plurality of SL PRS resource pools is a dedicated SL PRS pool.

36. The non-transitory computer-readable storage medium of claim **30**, the operations comprising:

decoding the DCI format to further determine a number of repetitions for the SL PRS transmission in the SL PRS resource pool.

37. The non-transitory computer-readable storage medium of claim **36**, the operations comprising:

decoding the DCI format to determine a resource pool index and a time gap value;

selecting the SL PRS resource pool as a shared SL PRS resource pool based on the resource pool index; and

determining a resource from the shared SL PRS resource pool for the SL PRS transmission based at least on the time gap value.

38. A user equipment (UE) configured for operation in a Fifth Generation New Radio (5G NR) network, the UE comprising:

front-end circuitry coupled to one or more antennas;

processing circuitry coupled to the front-end circuitry, the processing circuitry is to:

decode higher layer signaling received from a base station, the higher layer signaling to configure a plurality of sidelink positioning reference signal (SL PRS) resource pools;

decode a downlink control information (DCI) format received from the base station, the DCI format to configure scheduling information of a physical sidelink control channel (PSCCH) and a SL PRS, the scheduling information based on a SL PRS resource pool of the plurality of SL PRS resource pools;

decode SL control data associated with the PSCCH and the SL PRS, the SL control data and the SL PRS received from a second UE using the SL PRS resource pool; and

encode a second SL PRS for an SL PRS transmission to the second UE using the SL PRS resource pool; and

a memory coupled to the processing circuitry and configured to store the DCI format.

39. The UE of claim **38**, wherein the DCI format is a DCI format 3_0.

40. The UE of claim **39**, wherein the SL PRS resource pool of the plurality of SL PRS resource pools indicated by the DCI format 3_0 is a shared SL PRS pool.

* * * * *